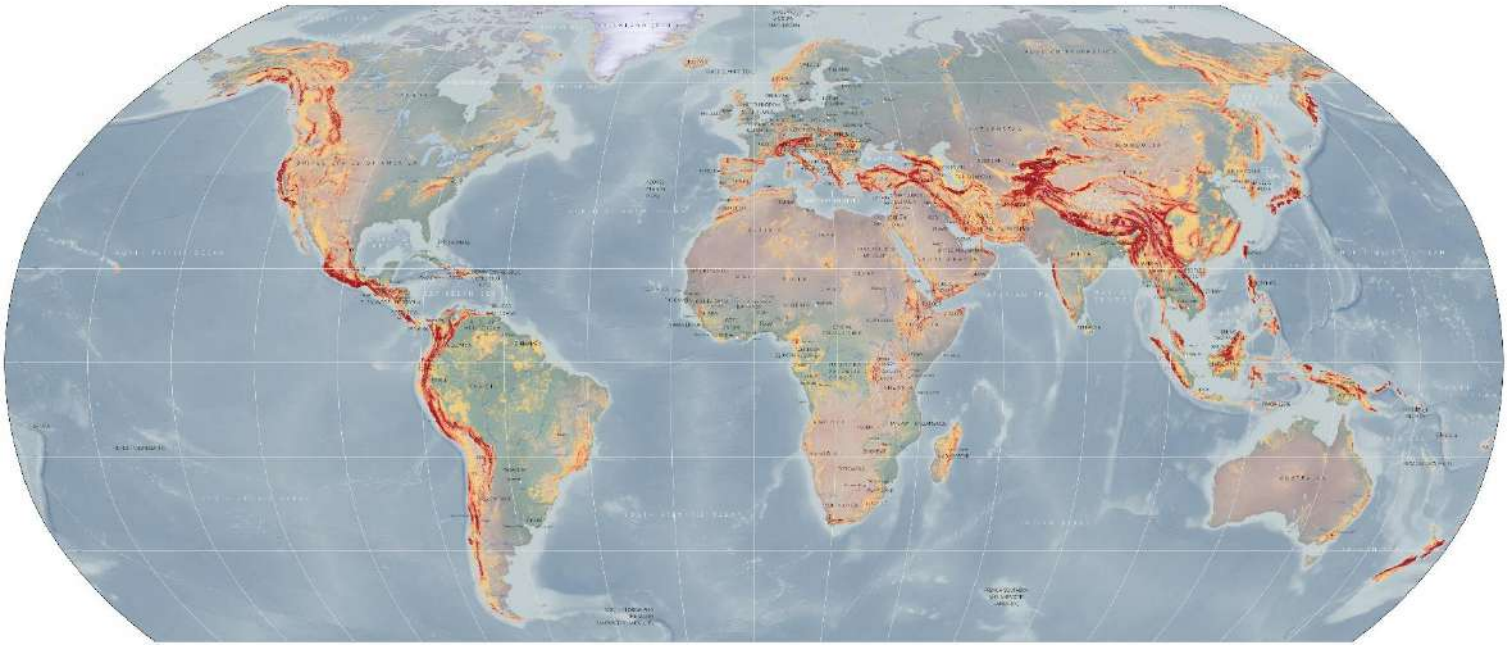
The World Bank

The Global Landslide Hazard Map

Final Project Report

Final Project Report | 26 June 2020



This report takes into account the particular instructions and requirements of our client. Although it is intended that the landslide hazard assessment can be used by the GFDRR and The World Bank to inform decision making at a regional scale, it is not intended for, and should not be relied upon by any third party and no responsibility is undertaken to any third party, for any specific use of the landslide hazard assessment.

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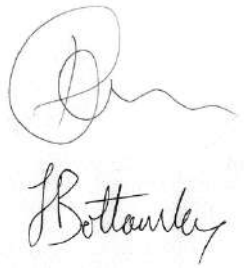
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Contents

		Page
1	**Introduction**	**4**
1.1	Background	4
1.2	Project Aims and Objectives	5
1.3	Hazard (and Risk) Framework	5
1.4	This Report	6
1.5	Definition of Terms and Acronyms	7
1.6	Accompanying Data	8
1.7	Limitations	9
1.8	Acknowledgements	9
2	**Global Landslide Hazard**	**10**
2.1	Introduction	10
2.2	Overview of Global Landslide Hazard	10
2.3	Landslide Hazard Assessments	12
3	**Landslide Hazard Assessment Methodology**	**16**
3.1	Introduction	16
3.2	Rainfall-Triggered Landslide Hazard Assessment	16
3.3	Earthquake-Triggered Landslide Hazard Assessment	25
3.4	Assessment of Landslide Seasonality	28
3.5	Inclusion of Hazard Data in ThinkHazard!	30
4	**Global Landslide Hazard Results**	**36**
4.1	Introduction	36
4.2	Global Overview of Landslide Hazard	36
4.3	Rainfall-Triggered Landslide Hazard (1980-2018)	38
4.4	Global Landslide Hazard Maps	40
5	**Discussion of Global Landslide Hazard Results**	**42**
5.1	Introduction	42
5.2	Comparison with Other Landslide Inventories	42
5.3	Comparison with Previous Studies	45
5.4	Effect of Large Storm Events	50
5.5	Significant Landslide Size	52
5.6	Error handling	55
6	**Summary and Conclusions**	**57**
6.1	Introduction	57
6.2	Summary of Landslide Hazard Results	57

6.3 What is new about *The Global Landslide Hazard Map* in comparison to existing maps? 58

7 Recommendations for Future Work 60

7.1 Introduction 60

7.2 Recommendations 60

8 References 61

Appendices

Appendix A

Global Landslide Hazard Maps

Appendix B

Tabulated Global Landslide Hazard Estimates

Appendix C

Summary of Qualitative Landslide Hazard Classification and Landslide Seasonality

Appendix D

Summary of Accompanying Data

Executive Summary

This report presents the findings of the Global Landslide Hazard Assessment Project undertaken by Ove Arup & Partners International Ltd (Arup) for the Global Facility for Disaster Reduction and Recovery (GFDRR) and The World Bank. Landslide information provided by NASA has been utilised in the landslide assessment and this important resource is gratefully acknowledged.

Landslide susceptibility describes the inherent properties of terrain which might make it more or less susceptible to failure e.g. geology, slope angle, elevation etc. For a landslide to initiate from a susceptible area of terrain a landslide trigger is required. This is typically rainfall but could also be ground shaking from an earthquake or related to human activity. An area can be of high landslide susceptibility but low landslide hazard if there is not the potential for a trigger of sufficient magnitude. Similarly, an area can be of low landslide susceptibility but high landslide hazard if there is the potential for a sufficiently large trigger. In order to better understand the frequency and potential impacts of landslides, it is therefore important to extend existing landslide susceptibility maps to present landslide hazard.

This project provides a systematic assessment of landslide hazard at a global scale. Landslides triggered by precipitation and landslides triggered by earthquakes have been determined separately and a combined qualitative landslide hazard assessment provided. The intention is that this landslide hazard information can be made available via the Think Hazard! online tool of the GFDRR and The World Bank so that decision makers around the world can access this data to inform land use, disaster risk management and investment decision making.

This project defines landslide hazard as the average annual frequency of occurrence of a significant landslide occurring within a defined area. The average time interval between each landslide event, or the recurrence interval (e.g. 1 in 100 years), is the inverse of the average annual frequency (e.g. 0.01). The landslide hazard calculations have been undertaken at a 1 sq. km grid globally and aggregated to different administration unit levels for summary and discussion purposes as required.

It is estimated that a total in the order of 15,500,000 significant rainfall-triggered landslides occurred worldwide between 1980 and 2018. The average annual number of significant rainfall-triggered landslides to occur globally is estimated to be in the order of 400,000 based on this analysis of data from 1980 to 2018. The average annual number of significant earthquake-triggered landslides is estimated to be in the order of 130,000.

Tabulated Landslide Hazard Results

Full tabulated estimates of landslide hazard by country are provided in Appendix B.

The first summary table presents the top ten highest rainfall-triggered landslide hazard countries by estimated average annual number of significant landslides. The second summary table presents the top ten highest earthquake-triggered landslide hazard countries by estimated average annual number of significant landslides.

Exec Table 1 – Estimated average annual number of significant rainfall-triggered landslides (1980-2018)

Country	Country code	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)
United States of America	259	36,150
China	147295	35,280
India	115	31,430
Philippines	196	23,110
Indonesia	116	22,220
Russian Federation	204	18,340
Myanmar	171	15,080
Brazil	37	13,360
Canada	46	11,780
Vietnam	264	11,490

Exec Table 3 – Estimated average annual number of significant earthquake-triggered landslides

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides
China	147295	20,950
Kyrgyz Republic	138	11,070
United States of America	259	10,710
Turkey	249	9,270
Mexico	162	7,510
Islamic Republic of Iran	117	6,580
Russian Federation	204	6,290
Tajikistan	239	5,810
Afghanistan	1	4,990
Nepal	175	4,550

Landslide Hazard Maps

Five new global-scale landslide hazard maps are presented in Appendix A of this report this report and in accompanying digital data format (details in Appendix D):

The Global Landslide Hazard Map: Qualitative Landslide Hazard

The Global Landslide Hazard Map: Qualitative Landslide Hazard presents a combined representation of global landslide hazard at a global scale. It is the combination of the sum of *The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)* and *The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard* which has then been simplified to four categories, ranging from Very low to High landslide hazard, based on the existing system used by ThinkHazard!. This map is also presented aggregated to admin. level 2 as **The Global Landslide Hazard Map: Admin. Level 2 Qualitative Landslide Hazard**.

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018)

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018) presents an estimate of rainfall-triggered landslide hazard which is representative for the period 1980-2018. The mean average landslide hazard values are in some parts of the world very clearly influenced by major rainfall/storm events which have occurred between 1980 and 2018. A variant of this map, **The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)** is also presented. The median average results appear to be less influenced by incidences of increased landslide hazard due to major rainfall/storm events which have occurred between 1980 and 2018.

The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard

The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard presents an estimate of earthquake-triggered landslide hazard at a global scale which builds upon work by Nadim et al. (2006; 2013) and uses the NASA Global Landslide Susceptibility Map as a base.

Landslide Seasonality

An assessment of landslide seasonality using a statistical analysis of the NASA Global Landslide Hazard Assessment for Situational Awareness (LHASA) has been undertaken. Landslide seasonality describes the months throughout the year when rainfall-triggered landslide hazard is highest due to seasonal weather patterns. The analysis shows distinct potential landslide seasonality trends for different geographies. The seasonality results for all countries are summarised in Appendix C.

1 Introduction

This document presents the final project report for the Global Landslide Hazard Assessment Project undertaken by Ove Arup & Partners International Ltd (Arup) for the Global Facility for Disaster Reduction and Recovery (GFDRR) and The World Bank.

1.1 Background

The GFDRR is a multi-donor partnership and grant-making financing mechanism. The facility supports on-the-ground technical assistance to help developing countries integrate disaster risk management (DRM) and climate change adaptation into development strategies, policies and investment programs. Between 2015 and 2018, the GFDRR implemented the development of new disaster risk information for multiple hazards across nine countries in Sub-Saharan Africa, contributing to the EU and ACP funded *Building Disaster Resilience to Natural Hazards in Sub-Saharan African Regions, Countries and Communities Program* for *Result Area 5 (R5), the Africa Disaster Risk Assessment and Financing Program*. The development of new disaster risk information was a necessary first step in improving the understanding of disaster risk in the region, providing fundamental inputs into disaster risk financing and insurance strategy development.

One component of the *R5* project focused on implementing national scale landslide hazard and risk assessments, with analysis led by Arup and supported by the British Geological Survey (BGS) (Arup-BGS, 2018). This component provided an innovative landslide risk assessment, which considered not only geographic variation in landslide susceptibility, but also considered factors which are known to trigger landslides (rainfall and seismic ground-shaking), leading to an estimation of landslide hazard. The development of these landslide hazard maps enabled an estimation of landslide frequency and size to be included in the assessment. This in turn enabled the estimation of building stock, infrastructure, and population that would coincide with potential landslide location, and to estimate risk in terms of annual frequency of impact on those exposures.

This Global Landslide Hazard Assessment project scales up the previous national-scale analysis and develops a landslide hazard map for the whole world. A global susceptibility map was recently published by NASA (Stanley and Kirschbaum, 2017), and this is used as a base for the development of hazard estimates. The hazard maps incorporate landslide triggers from global rainfall datasets and seismic hazard maps. The outputs of the project provide a route to estimating risk from landslides to building stock, infrastructure and population across the world, and will improve derivative products such as the classification of landslide hazard in ThinkHazard! (www.thinkhazard.org), which provides free and open online access to hazard information and risk reduction guidance.

1.2 Project Aims and Objectives

Aims

The aim of this project is to develop a global-scale landslide hazard map. This map will then be used to generate improved global coverage for landslide hazard levels on ThinkHazard!, the GFDRR web portal for global natural hazard information.

Objectives

The three main objectives of this project are:

- To develop a global-scale landslide hazard map with landslide hazard scaled in terms of probability or frequency of landslide occurrence;
- Summarise these quantitative estimates of landslide hazard using qualitative terminology and data which are aggregated to administrative unit levels 0, 1 and 2 for inclusion in ThinkHazard!; and
- Undertake an assessment of landslide seasonality based on a historical analysis of NASA Global Landslide Hazard Nowcasts.

1.3 Hazard (and Risk) Framework

Corominas et al. (2014) describe a comprehensive landslide hazard and risk framework comprising the following key stages:

- Hazard identification;
- Hazard assessment;
- Inventory of elements at risk;
- Vulnerability assessment of elements at risk; and
- Risk estimation.

Hazard identification deals with using existing literature, landslide inventories and other sources to characterise the nature of landslide hazards within a given area. Hazard assessment uses observations of the landslide hazard to interpret the susceptibility of the study area terrain to generate further landslides and characterises the magnitude and frequency of these hazards. By then overlaying the exposure analysis (which is essentially an inventory of elements at risk such as buildings, people, roads etc.) with the hazard maps it is possible to identify those elements which are exposed to some level of hazard. Then, typically using expert opinion, empirical data and physically based models, it is possible to determine how vulnerable the elements at risk are to landslides. Finally, risk estimation is the culmination of the risk analysis process and presents landslide risk to the exposed people or assets, typically using an annualised or probabilistic metric e.g. average annual landslide risk to buildings or total financial losses occurring within a given return period.

This global landslide hazard assessment project addresses the hazard assessment component of the hazard and risk framework. This project uses the NASA Global Landslide Susceptibility Map and develops this into a global landslide hazard assessment which considers rainfall and earthquake-triggering. Exposure, vulnerability and risk estimation are not included as part of this project.

Why develop a global landslide hazard map when a global landslide susceptibility map is already available?

Landslide susceptibility describes the inherent properties of the terrain which might make it more or less susceptible to failure e.g. geology, slope angle, elevation etc. For a landslide to initiate from a susceptible area of terrain a landslide trigger is required. This is typically rainfall but could also be ground shaking from an earthquake or related to human activity. If a landslide is triggered then it is referred to as a landslide hazard and will have landslide hazard metrics related to size, velocity and frequency of occurrence. An area can be of high landslide susceptibility but low landslide hazard if there is not the potential for a trigger of sufficient magnitude. Similarly, an area can be of low landslide susceptibility but high landslide hazard if there is the potential for a sufficiently large trigger. In order to better understand the frequency and potential impacts of landslides, it is therefore important to extend from susceptibility to hazard.

1.4 This Report

This report follows the following general structure:

- Section 1: This section contains a general introduction to the project, the origins of the work and the aims and objectives. This section also includes a summary of the maps and data which accompany this report.
- Section 2: The second section contains an overview of global landslide hazard and previous global and large-scale landslide hazard assessments.
- Section 3: The methodology used for the global landslide hazard assessment is presented, including details about how the data are prepared for inclusion in ThinkHazard!.
- Section 4: Results from the global landslide hazard assessment are presented, including tabulated country summaries and time-series plots showing the estimated variation in rainfall-triggered landslide hazard between 1980 and 2018.
- Section 5: This section presents a discussion on the results from the global landslide hazard assessment.
- Section 6: This section presents a summary of the key findings and conclusions from this project.
- Section 7: The methods developed by this project and the results discussed herein are an exciting starting point for further developments in characterising global landslide hazard. This section presents an overview of opportunities for developing further work.

1.5 Definition of Terms and Acronyms

1.5.1 Definition of Terms

Antecedent Rainfall

Antecedent rainfall describes the sum of rainfall to occur over a given duration prior to a specified day. For example, the 3-day antecedent rainfall for 15/01/2020 is defined as the sum of rainfall from 13/01/2020, 14/01/2020 and 15/01/2020. Some use-cases may choose to include the specified day as part of the antecedent period whilst others may omit it. This project includes the specified day in the stated antecedence period.

Landslide Hazard

Cruden (1991) defines a landslide as *the movement of a mass of rock, earth or debris down a slope*, with the qualification that landslides are not confined to the land, nor to simple sliding failure. Landslides can encompass falls, flows, topples and spreads in addition to slides (Varnes, 1978) and have been recognized on the continental shelf (Twichell et al., 2009), on the sea floor (Prior and Hooper, 1999) and even on Mars (Quantin et al., 2004). Landslide hazards may be single, sequential or combined in their origin and effects and can be characterized by location, intensity or magnitude, frequency and probability (UNDRR, 2017).

Landslide Inventory/Catalogue

Landslides inventories and the preparation of landslide inventory maps is the simplest form of landslide mapping. A landslide inventory records the location, date of occurrence and types of landslides which have occurred within a specified area. There are broadly two types of landslide inventory: landslide event inventories, which contain all of the landslides which can be directly attributed to a particular trigger event; and historical landslide inventories, which represent the sum of landslide activity over a longer period of time (Malamud et al., 2004). The NASA Global Landslide Catalogue is a historical landslide inventory.

Landslide Risk

Landslide risk can be defined as a measure of the probability and severity of an adverse effect to health, property or the environment as a result of landslides (Corominas et al., 2014).

Landslide Susceptibility

Landslide susceptibility describes the inherent properties of the terrain which might make it more or less susceptible to failure e.g. geology, slope angle, elevation etc.

Landslide Trigger

For a landslide to initiate from a susceptible area of terrain a landslide trigger is required. This is typically rainfall but could also be ground shaking from an

earthquake or related to snowmelt, human activity or some other direct cause of failure initiation.

Logistic Regression

Logistic regression is a kind of statistical analysis that is used to predict the outcome of a dependent variable based on prior observations. For example, a logistic regression algorithm might be used to determine the winner of a presidential election based on past election results and economic data (Techopedia, 2020). Logistic regression is the appropriate regression analysis to conduct when the dependent variable is dichotomous (binary) – elected or not elected; landslide or no landslide.

ThinkHazard!

ThinkHazard! (http://thinkhazard.org/en/) is a GFDRR-managed resource which provides a general view of the natural hazards, for a given location, that should be considered in project design and implementation to promote disaster and climate resilience. The tool highlights the likelihood of different natural hazards affecting project areas (very low, low, medium and high), provides guidance on how to reduce the impact of these hazards, and where to find more information. The hazard levels provided are based on published hazard data, provided by a range of private, academic and public organizations.

1.5.2 Acronyms

BGS

British Geological Survey

GFDRR

Global Facility for Disaster Reduction and Recovery

GSHAP

Global Seismic Hazard Assessment Programme

NASA

The National Aeronautics and Space Administration

NOAA

National Oceanic and Atmospheric Administration

1.6 Accompanying Data

This report is accompanied by several datasets. Metadata for these datasets are presented are Appendix D.

1.7 Limitations

General

This report takes into account the particular instructions and requirements of our client. Although it is intended that the landslide hazard assessment can be used by the GFDRR and The World Bank to inform decision making at a regional scale, it is not intended for, and should not be relied upon by any third party and no responsibility is undertaken to any third party, for any specific use of the landslide hazard assessment.

Precision

Tabulated estimates of average annual number of significant landslides are reported to the nearest ten. Tabulated estimates of average annual number of significant landslides are reported to four decimal places. Data accompanying this reporting are unlimited and are typically reported to a higher precision.

Accuracy

It is important to note that these data represent average estimated values calculated at a global scale using global-scale datasets. Expert judgement should be used when interpreting and analysing these values, with due consideration given to the high-level nature of this study. The reported precision of results is not intended to convey any particular accuracy, but rather to show spatial and temporal variation.

1.8 Acknowledgements

Arup wish to acknowledge the valuable contributions made by The World Bank project team led by Dr Stuart Fraser and Dr Mattia Amadio (World Bank). We also thank Dr Dalia Kirschbaum and Dr Robert Emberson (NASA) for their support at all stages of this project, from inception through to attendance at review workshops.

2 Global Landslide Hazard

2.1 Introduction

This section of the report provides a very high-level overview of global landslide hazard and hazard assessments. It should be noted that there is a vast range of literature related to landslide hazard. This section aims to touch upon just some of the key references which relate directly to specific aspects of this project.

2.2 Overview of Global Landslide Hazard

Why study global landslide hazard?

The annual rate of reported fatalities resulting from landslides worldwide is, generally speaking, increasing (Petley, 2011). Petley, Dunning and Rosser (2005) cite a reported 106,622 global landslide-related fatalities between 1980 and 2000. Petley (2012) reports a further 80,058 fatalities resulting from landslides between 2004 and 2010. Furthermore, these estimates are likely to underestimate the true loss of life, partly because some landslide events, particularly in remote parts of the World, are poorly reported or recorded (Sepúlveda and Petley, 2015). In addition to this, large-scale landslide-triggering events are capable of causing near-simultaneous and devastating loss of life. Landslides resulting from the 2015 Nepal earthquakes are estimated to have caused more than 8,500 fatalities (Frank and Gordon, 2015). More than 20,000 fatalities are estimated to have resulted from landslides which were triggered by the 2008 Wenchuan earthquake in China (Petley, 2013).

The built environment is particularly exposed to the potential for damage and loss resulting from landslides, driven in-part by rapid global urbanization. As cities swell to accommodate new inhabitants, city-limits spread into new areas. In the developing world in particular, due to the pressure to maintain an urban-centric livelihood, villages, towns, districts – whole neighbourhoods and communities bloom in areas which might previously have been considered unfavourable or at least unpreferable for construction. If, as is often the case, this urbanization is not accompanied by the requisite master-planning or regulation, slope management practices are often neglected and hence landslide susceptibility can increase. Surface runoff increases due to insufficient drainage; trees are cut down for fuel and to make space for development; and slopes are carved open to make way for buildings and roads. The Regent Landslide in Freetown, Sierra Leone, is a devastating example of what can happen when landslides occur in urban areas in the developing world. The August 2017 landslide claimed ~1,000 lives and caused ~30M USD damage and loss, the majority of which was to buildings and infrastructure (World Bank, 2017; Redshaw et al., 2019).

Gaining a better understanding of the frequency and magnitude of landslide hazards at a global scale allows us to better plan for, manage and mitigate the potential effects of landslides on people and the built environment. Whilst many in-depth and detailed landslide hazard studies have been undertaken for the developed world, many parts of the developing world are without such a resource.

By undertaking a global-scale study, we provide high-level quantitative landslide hazard information which can be used for broad planning or hazard studies for anywhere in the world.

Where is landslide hazard highest?

As a very basic rule-of-thumb, landslide hazard is most prominent in mountainous areas – the Himalayas, the Alps, the Andes, the Rockies etc. Nadim et al. (2006; 2013) cite high hazard areas including: Central America, north-western South America, north-western USA and Canada, the Caucasus region, the Alborz and Zagros mountain ranges in Iran, Turkey, Tajikistan, the Kyrgyz Republic, the Himalayan belt, Taiwan, the Philippines, Indonesia, New Zealand, New Guinea, Italy and Japan.

What types of landslide are there?

Cruden (1991) defines a landslide as *the movement of a mass of rock, earth or debris down a slope*, with the qualification that landslides are not confined to the land, nor to simple sliding failure. Landslides can encompass falls, flows, topples and spreads in addition to slides (Varnes, 1978) and have been recognized on the continental shelf (Twichell et al., 2009), on the sea floor (Prior and Hooper, 1999) and even on Mars (Quantin et al., 2004). They can occur at a variety of rates ranging from the near indiscernibly slow (mm's/year) to the almost incomprehensibly rapid (10s – 100s km/hour), and at different scales ranging from small localised slumps (<1 m^3) to island-scale collapses (>100,000,000 m^3) (Weidinger et al., 2014).

A necessary assumption for undertaking a study such as this at a global-scale is that the detail of landslide mechanism be disregarded, and that landslides are thought of more broadly by Cruden's simple definition as *the movement of a mass of rock, earth or debris down a slope*. This assumption is discussed and further justified in Section 3.2.6.

Triggering of landslides

The two main triggers of landslides are rainfall, and in areas of seismic hazard, ground shaking. Rainfall provides a regular, seasonal control to landslide occurrence, whilst seismicity provides a much more unpredictable and potentially widespread source of landslide hazards.

Snowmelt can also trigger landslides, as can anthropogenic causes such as road cuttings and embankments. Particularly in the past few decades or so, fatalities associated with mining-related landslides have increased, reflecting increased mining activity, particularly in Asia (Froude and Petley, 2018).

Rainfall and earthquake triggering are considered for this project.

What effect will climate change have on landslide hazards?

It is broadly accepted that with climate change, rainfall events are likely to become more seasonal and more intense – periods of drought and subsequent flooding will become more pronounced. According to Zhang et al. (2013),

attributed intensification of annual maximum 1-day precipitation is about 5.2% per degree of warming (uncertainty range 1.5 - 9.3%). This will have the effect of locally increasing landslide hazard. This is because during periods of drought, depending on the type of soils and land cover, the ground can dry up, forming desiccation cracks. During later deluges of rain, these cracks allow rainwater to rapidly infiltrate the ground, increasing groundwater levels and causing slope instability. Landslide hazard will also potentially increase in areas subject to a net increase in total rainfall. It should also be noted that increased temperature and precipitation can potentially result in increased vegetation growth, which may reduce landslide hazard in certain scenarios.

2.3 Landslide Hazard Assessments

2.3.1 Landslide Inventories

Landslide inventories form the basis for landslide hazard assessments. They provide information on the spatial distribution, timing, magnitude and trigger of landslide events, and can be broadly sub-divided into two main types (Malamud et al., 2004):

- Landslide event inventories, whereby the records are associated with a specific triggering event (usually either a storm/rainfall event or an earthquake); or
- Historical landslide inventories, which represent the sum of many landslide-triggering events over a period of time.

Historical landslide inventories are notoriously incomplete. This is because landslide events cannot be accurately logged without direct observation, either by field mapping or more commonly using remotely sensed data[1]. Moreover, when logging historical landslides, it is possible to mis-identify or miss altogether smaller landslides, evidence for which may have become obscured throughout recent geological history. It is also difficult to date historical events precisely, and unless they occurred within the last half-century or so, to understand the rainfall conditions which preceded them.

NASA Global Landslide Catalogue

The Cooperative Open Online Landslide Repository (COOLR) includes NASAs original Global Landslide Catalogue; landslides reported in the Landslide Reporter Catalogue and collated landslide inventories by other institutions. This catalogue is the largest available global-scale landslide catalogue and includes more than 13,000 events (as of November 2019) dating back to 1957. The catalogue includes information on date of occurrence, location, location accuracy, landslide type, trigger, size, setting and information on the number of fatalities or injuries caused.

[1] In certain circumstances landslides can be identified by their seismic signal e.g. Zhang and He (2019).

The Global Fatal Landslide Database

The Global Fatal Landslide Database contains records of non-seismic fatal landslides to have occurred worldwide since 2002. The most recent write-up of the database (Froude and Petley, 2018) provides analysis for events between 2004 and 2016. The data show a total of 55,996 fatalities in nearly 5,000 distinct landslide events. Froude and Petley note that Asia represents the dominant geographical area for fatal landslides and that landslide occurrences triggered by human activity are broadly increasing.

2.3.2 Global Hazard Assessments

Global Landslide Hazard Distribution v1

Collaboration between the Columbia University Center for Hazards and Risk Research (CHRR), the Norwegian Geotechnical Institute (NGI) and the Columbia University Center for International Earth Science and Information Network (CIESIN) produced a global landslide hazard distribution based on a qualitative assessment of a range of susceptibility factors, including slope, soil, soil moisture conditions, precipitation and seismicity (CHRR, NGI and CIESIN, 2005; Dilley et al., 2005).

Assessment of Global Landslide Hazard Hotspots

Nadim et al. (2006) present a global scale landslide hazard analysis based on global datasets of climate, lithology, earthquake activity and topography (later revised by Nadim et al., 2013). The qualitative assessment classifies and scores the different susceptibility and triggering factors and uses locations where good landslide inventories are available to spot-check hazard estimates. Nadim et al. also propose approximate annual frequencies associated with each of the qualitative hazard rankings. Risk is then estimated by comparison with exposure datasets and using some high-level assumptions, for example:

"It was assumed that a typical serious landslide or avalanche event affects, on average, 10% of the population living in the spatial unit used for analyses (~1 km^2)."

Overall, Nadim et al. show that fairly good agreement can be found between their global hazard model and the inventories against which it was compared. Furthermore, Nadim et al.'s estimates of risk (average annual fatalities of ~4,300) seem plausible and in-line with other global records. This serves to highlight the value of high-level studies such as those by Nadim et al. or this study – it isn't always necessary to be 100% accurate, particularly when doing a first pass assessment. For global studies, understating the approximate hazard levels to an order of magnitude or so is extremely useful.

The data from Nadim et al.'s global landslide hazard hotspots study is readily available using the UN Global Risk Data Platform[2].

[2] https://preview.grid.unep.ch/

NASA Global Landslide Susceptibility Map

The NASA Global Landslide Susceptibility Map (Stanley and Kirschbaum, 2017) was developed to assess the underlying landslide susceptibility of terrain. This includes quantitative information on whether roads have been built, trees have been cut down or burned, a major tectonic fault is nearby, the local bedrock is weak, and/or the hillsides are steep.

For this project an advanced copy of a revised NASA Global Landslide Susceptibility Map was provided by Robert Emberson (NASA) on 09/10/2019. The current publicly available data are available to download from https://pmm.nasa.gov/applications/global-landslide-model.

NASA Global Landslide Hazard Assessment for Situational Awareness (LHASA)

The NASA-developed LHASA uses the 30-minute temporal resolution Global Precipitation Measurement (GPM) Integrated Multi-satellitE Retrievals for GPM (IMERG) data to provide an up-to-date global estimate of landslide hazard (Kirschbaum, Stanley and Simmons, 2015; Kirschbaum, Stanley and Zhou, 2015; Kirschbaum, Stanley and Yatheendradas, 2016). A LHASA landslide *nowcast* can be created by comparing GPM data from the preceding seven days to the long-term precipitation record provided by the Tropical Rainfall Measuring Mission (TRMM) Multi-satellite Precipitation Analysis (TMPA). In places where the precipitation is unusually high, the susceptibility of the terrain is queried using the NASA Global Landslide Susceptibility Map. If the terrain is classified as moderate landslide susceptibility, then a *moderate-hazard nowcast* is issued and if the terrain is classified as high landslide susceptibility then a *high-hazard nowcast* is issued.

2.3.3 Large-Scale Regional Hazard Assessments

National-Level Landslide Risk Profiles for Sub-Saharan Africa

As part of the *R5* project (introduced in Section 1.1) Arup and the BGS developed country-scale landslide hazard and risk information for nine countries in Sub-Saharan Africa, namely: Ethiopia; Kenya; Uganda; Niger; Senegal; Malawi; Mali; Mozambique and Cabo Verde (Redshaw et al., 2017; Arup-BGS, 2018). The project used susceptibility data including soil type, elevation, slope angle etc. in conjunction with global rainfall and seismic hazard datasets to produce separate rainfall and earthquake-triggered landslide hazard estimates. The project also developed these through to an assessment of risk, providing estimates of average annual losses to population, buildings, infrastructure, critical facilities and GDP. Using global climate models, the project produced estimates of potential future (2050) hazard and risk in each of the nine countries.

Generation of a Landslide Risk Index Map for Cuba using Spatial Multi-Criteria Evaluation

Castellanos Abella and Van Westen (2007) produced a semi-quantitative landslide hazard and risk model for Cuba using spatial multi-criteria evaluation methods.

They separated *hazard indicators* into susceptibility factors (or as they call *conditioning factors*) and triggering factors, using slope angle, land use, geology, soil, geomorphology, slope length, drainage density and internal relief to characterise susceptibility and rainfall and seismicity to account for triggering. Because the available landslide inventory was not suitably complete for numerically-driven calibration, Castellanos Abella and Van Westen (2007) relied upon expert opinion. They note that:

"Although this method is subjective, it allows the incorporation of expert opinion and the use of group decision making and therefore is leading to reliable results, given the scale."

The study provides a good basis upon which to rank provinces and municipalities in order of importance for more detailed studies and landslide risk reduction methods.

Caribbean Handbook on Risk Information Management

The Caribbean Handbook on Risk Information Management (CHARIM[3]) provides an extremely well documented resource of information for natural hazard and risk management in Belize, Dominica, St. Lucia, St. Vincent and the Grenadines and Grenada. One aspect of CHARIM was the production of new landslide susceptibility and hazard information. This was achieved using spatial multi-criteria evaluation techniques in comparison to available landslide inventories. A total of 16 susceptibility datasets were considered for the analysis, including topographic, drainage, geological, soil and land cover type factors. For Saint Vincent, CHARIM concluded that ~42% of the total land area was of low landslide susceptibility, ~32% was of moderate landslide susceptibility and ~26% was of high landslide susceptibility (Van Westen, 2016).

Identification of Landslide Hazard and Risk 'Hotspots' in Europe

Jaedicke et al. (2014) present a study whereby two independent landslide hazard models were developed for Europe using largely the same input data. One model was heuristic and led by expert judgement, whilst the other was more statistical, using a logistic regression-based approach. Both models well identify areas of high landslide hazard for shallow landslides and rock falls, but do not well identify slow moving areas containing slow moving landslide hazards. Jaedicke et al. (2014) conclude that small Alpine countries such as Lichtenstein have highest relative exposure to landslide hazards, whilst Italy was found to have the highest absolute exposed population (risk). It is estimated that 1.3 – 3.6M people live in areas of high landslide hazard and that 9,000 – 20,000 km of roads and railways are exposed to high landslide hazards.

[3] http://www.charim.net/

3 Landslide Hazard Assessment Methodology

3.1 Introduction

This section of the report describes the methodology used for the global landslide hazard assessment. One of the aims of this project is to extend the existing NASA Global Landslide Susceptibility Map to include some component of hazard; some appreciation of the frequency and magnitude with which landslides occur around the world. Landslide catalogues are known to be significantly incomplete. If a landslide happens in a remote area there is little chance it will be reported and hence little chance it will be recorded in any landslide inventory. This makes the use of landslide inventories for creating hazard maps problematic, particularly at large scales. This project proposes a sensible methodology by which to correct for this incompleteness and to give a reasonable approximation of rainfall-triggered landslide hazard based on the data available, supported by expert judgement where necessary. A different method is proposed by which earthquake-triggered landslide hazard can be estimated.

3.2 Rainfall-Triggered Landslide Hazard Assessment

3.2.1 General Approach

The rainfall-triggered landslide hazard assessment is done by undertaking an assessment of the rainfall conditions which occurred in the run-up to the triggering of landslides recorded in the NASA Global Landslide Catalogue. Because landslide inventories are notoriously incomplete, only landslides which are recorded as having occurred in populated places and in countries where the primary language is English (and hence the events are more likely to be picked up and incorporated into the NASA Global Landslide Catalogue) are used to develop the relationship between *what sort of rainfall conditions might cause landslides* and *what sort of rainfall conditions might not*. This relationship is then applied globally to areas of any or no population to give an estimate of global landslide hazard.

Figure 1 summarises the general approach for the estimation of quantitative rainfall-triggered landslide hazard.



Figure 1 – Overview of the general approach for the estimation of quantitative rainfall-triggered landslide hazard.

3.2.2 Data Sources

In addition to the NASA Global Landslide Catalogue and the NASA Global Landslide Susceptibility Map (introduced in Section 2.3), the following data sources were used for the rainfall-triggered landslide hazard assessment.

NOAA CPC Rainfall Data

Rainfall data were sourced from the US National Oceanic and Atmospheric Administration (NOAA) Climate Prediction Center (CPC) (NOAA CPC, 2019). The data are gridded daily rainfall totals at a spatial resolution of 0.5 degrees (~50km). The data are available from 01/01/1979 to the present day, updated daily.

WorldPop Population Data

WorldPop Population Data (WorldPop, 2018) present the total estimated number of people per grid-cell at a spatial resolution of ~1 km. Population estimates refer to 2018.

3.2.3 Pre-processing

NASA Global Landslide Catalogue

The NASA Global Landslide Catalogue was pre-processed prior to use in the rainfall-triggered landslide hazard assessment.

- Only records of landslides with a reported accuracy of 1 km or better were used. This step is required to ensure that the rainfall and landslide susceptibility characteristics are accurately sampled at the location of the landslide event.
- Only records of landslides with a reported rainfall trigger were used.
- Only records of landslides which occurred in places populated by more than 500 people/sq.km were used. This is done to increase the quality of the dataset in terms of it's potential for identifying the sorts of rainfall conditions which might cause landslides and the sorts of rainfall conditions which might not. The inventory is more likely to be more complete in places of higher population.
- Only records of landslides which occurred in countries where English is the primary language were retained. This is because it is well known that landslides which occur in countries where English is not the primary language are under-represented in the NASA Global Landslide Catalogue. This is due to the way in which the catalogue is populated using English language-based search terms.
- Only landslides reported between 2010 and 2018 were retained. This is because during this period the number of landslides reported in each year is high and consistent (Figure 2). We acknowledge that this is a relatively modest time-frame when undertaking an assessment of natural processes.

NOAA CPC Rainfall Data

The NOAA CPC Rainfall Data were pre-processed prior to inclusion in the rainfall-triggered landslide hazard assessment.

- Rainfall data were cropped to the spatial extent of the NASA Global Landslide Susceptibility Map.
- In areas where a given date was missing a value for a small number of years in a relatively small area, but data were available for most years in that area, the average value from other years was applied.
- Around the perimeter and at the northern and southern extremities of the NASA Global Landslide Susceptibility Map some rainfall values were missing. These were filled from the nearest adjacent areas up to a maximum distance of 200 km.
- Areas missing values which were greater than 200 km from the nearest observation were filled with the mean rainfall value for the corresponding latitude. This only really applies to a handful of small atolls in the Pacific Ocean.

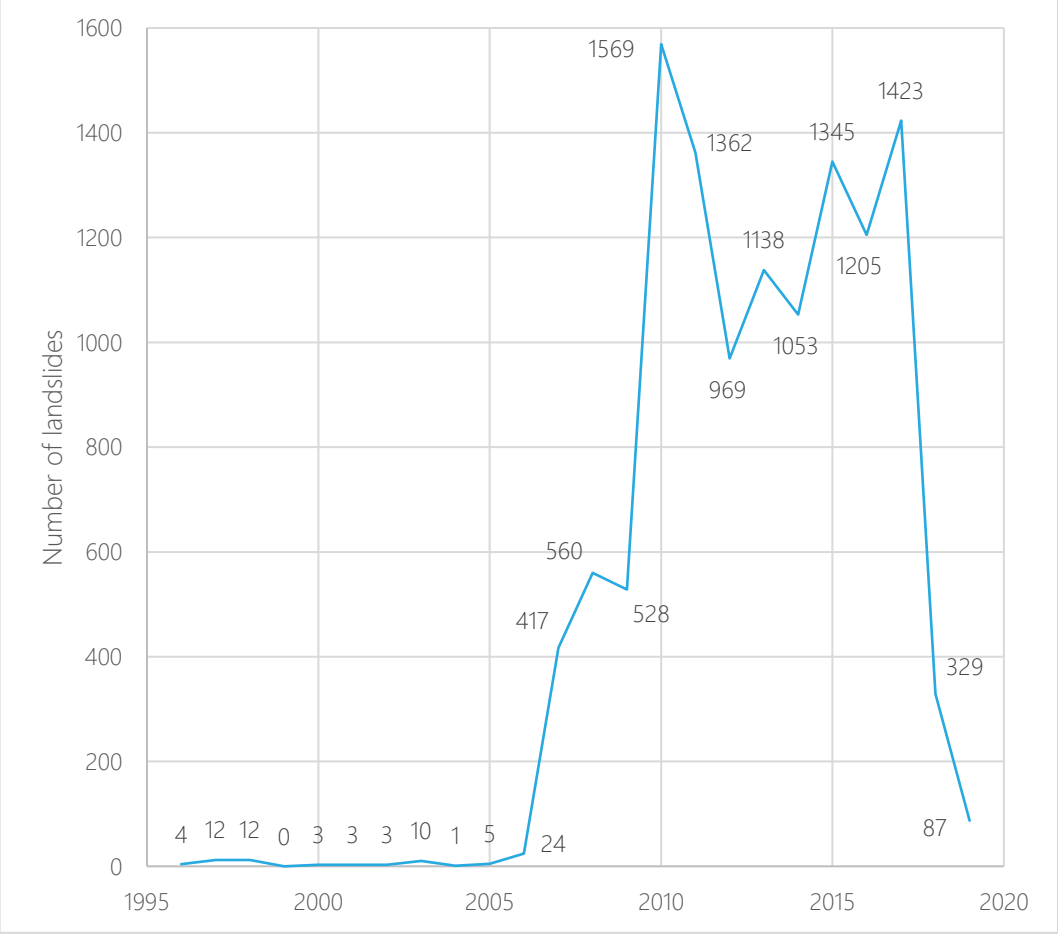


Figure 2 – Number of landslides reported each year in the NASA Global Landslide Catalogue (1995 – 2019 only)

- Antecedent rainfall was calculated at intervals of 5, 10, 15, 20 and 25 days for each date between 1980 and 2018.
- Following completion of the rainfall-triggered landslide hazard calculations it was identified that some, very limited anomalous rainfall values remained in the NOAA CPC Rainfall Data. Details of how these anomalous rainfall values and the correspondingly anomalous landslide hazard estimations were excluded from the final datasets are included in Section 5.6.

3.2.4 Rainfall-Triggered Landslide Hazard Data Modelling

Combined Dataset

To calibrate a model to look at the relationship between rainfall and landslide occurrence it was first necessarily to compile a combined dataset. This combined dataset contains examples of both the types of rainfall and susceptibility patterns which were observed to have caused landslides, and those which were observed to have not caused landslides. The dataset contains the following fields:

- Cell ID: A unique identifier for each grid square covering the world at the same spatial resolution, coverage and origin as the NASA Global Landslide Susceptibility Map.
- Date: The date to which the rainfall conditions refer.
- NASA Global Landslide Susceptibility Map ranking: The susceptibility ranking for that cell, which is represented by a score of 1 – 5, which corresponds to low – high susceptibility.
- Rainfall on the day: The rainfall which occurred on the day specified in the *Date* field in the cell specified in the *Cell ID* field.
- Cumulative rainfall over the previous N days, where N = 5, 10, 15, 20 and 25 days: This is the sum of the rainfall over the days leading up to the day specified in the *Date* field in the cell specified in the *Cell ID* field. The date itself is included in this antecedence period. For example, for N = 5 for a date of 10/10/2018, the reported value would be the sum of rainfall to have occurred on 06/10/2018, 07/10/2018, 08/10/2018, 09/10/2018 and 10/10/2018.
- Landslide occurrence: This is a binary field which indicates if a landslide was reported in that cell on that date (1) or not (0).

Table 1 presents an example of the compiled dataset.

Table 1 – Example of the compiled rainfall and landslide dataset

Cell ID	Date	NASA Global Landslide Susceptibility Map ranking	Rainfall on the day (mm)	Cumulative rainfall over the previous N days (mm)					Landslide occurrence
				N = 5	N = 10	N = 15	N = 20	N = 25	
1	01/01/2015	1	0	0	0	0	0	0	0
2	01/01/2015	1	10	22	60	150	200	220	1
3	21/09/2017	4	5	5	5	5	5	6	0
4	30/12/2018	5	70	90	95	110	270	350	1

All the sub-sampled landslide catalogue was included in the compiled dataset (i.e. all the 1s). It was necessary to sub-sample the non-event data (0s) due to the high number of days per cell where no landslide occurred. 5,000 randomly selected non-event cell IDs and dates were selected from each of the five susceptibility ranking classes – 25,000 in total.

Rainfall-Triggered Landslide Hazard Modelling Approach – Machine Learning – Logistic Regression

The aim here is to apply a modelling methodology which can identify and predict the probability of a landslide (landslide occurrence = 1) given the susceptibility ranking, the on-the-day and the antecedent rainfall conditions.

Chandrayan (2019) defines logistic regression as:

"… a statistical method for analysing a dataset in which there are one or more independent variables that determine an outcome. The outcome is measured with a dichotomous variable (in which there are only two possible outcomes). It is used to predict a binary outcome (1/0, Yes/No, True/False) given a set of independent variables."

Chandrayan (2019) goes on to note:

"… in simple words, it predicts the probability of occurrence of an event by fitting data to a logit [logistic regression] function."

Because the combined dataset is not well balanced (more 0s than 1s; more non-events than events); the non-events (0s) are sub-sampled from the full population of non-events; and there are significantly fewer events (1s) than non-events (0s) in the full population, modelling for this project makes use of a particular branch of logistic regression referred to as *rare events logistic regression* (King and Zhen, 2001). Rare events logistic regression problems are those where the variable to be predicted is rare in its occurrence in comparison to the frequency of non-events. King and Zhen (2001) begin their much-cited formative paper on rare events:

"We address problems in the statistical analysis of rare events data – binary dependent variables with dozens to thousands of times fewer ones (events, such as wars, coups,

presidential vetoes, decisions of citizens to run for political office, or infections by uncommon diseases) than zeros (non-events)."

Likewise, there are many, many times more days where rainfall occurred and did not cause a landslide, than times where it did.

Rainfall-Triggered Landslide Hazard Model Calibration

Five logistic regression models were developed for this project – one for each of the five susceptibility classes from the NASA Global Landslide Susceptibility Map. Developing one model for each class allowed the rare events correction to be applied on a class-by-class basis and hence better represent the probability of landslide occurrence in each susceptibility class based on different rainfall conditions.

The data were randomly split into a training (80%) and test (20%) dataset to allow later verification of the model accuracy.

Rainfall-Triggered Landslide Hazard Model Verification

Figure 3 shows the success rate and prediction rate curves for both the training and test data for each of the five models defined for this project (one for each landslide susceptibility class)[4].

Table 2 shows the area under curve values for each of the success rate and prediction rate curves for each landslide susceptibility class model.

Table 2 – Area under curve values for the success rate and prediction rate curves for each landslide susceptibility class model

Susceptibility Class	Success rate	Prediction rate
1 (Very Low)	0.81	0.87
2	0.85	0.84
3	0.85	0.86
4	0.90	0.83
5 (Very High)	0.85	0.83

Rasyid et al. (2016) classify the accuracy of a diagnostic test such as this using the ranges 0.50 to 0.60 (fail), 0.60–0.70 (poor), 0.70–0.80 (fair), 0.80–0.90 (good), and 0.90–1.00 (excellent). According to this classification the models developed for this project therefore show good accuracy.

[4] Success and prediction rate curves are widely used, particularly in landslide hazard assessment, to quantify the goodness-of-fit of a numerical model. Examples include Mezughi et al (2011); Deng et al. (2017); and Rasyid et al. (2016).

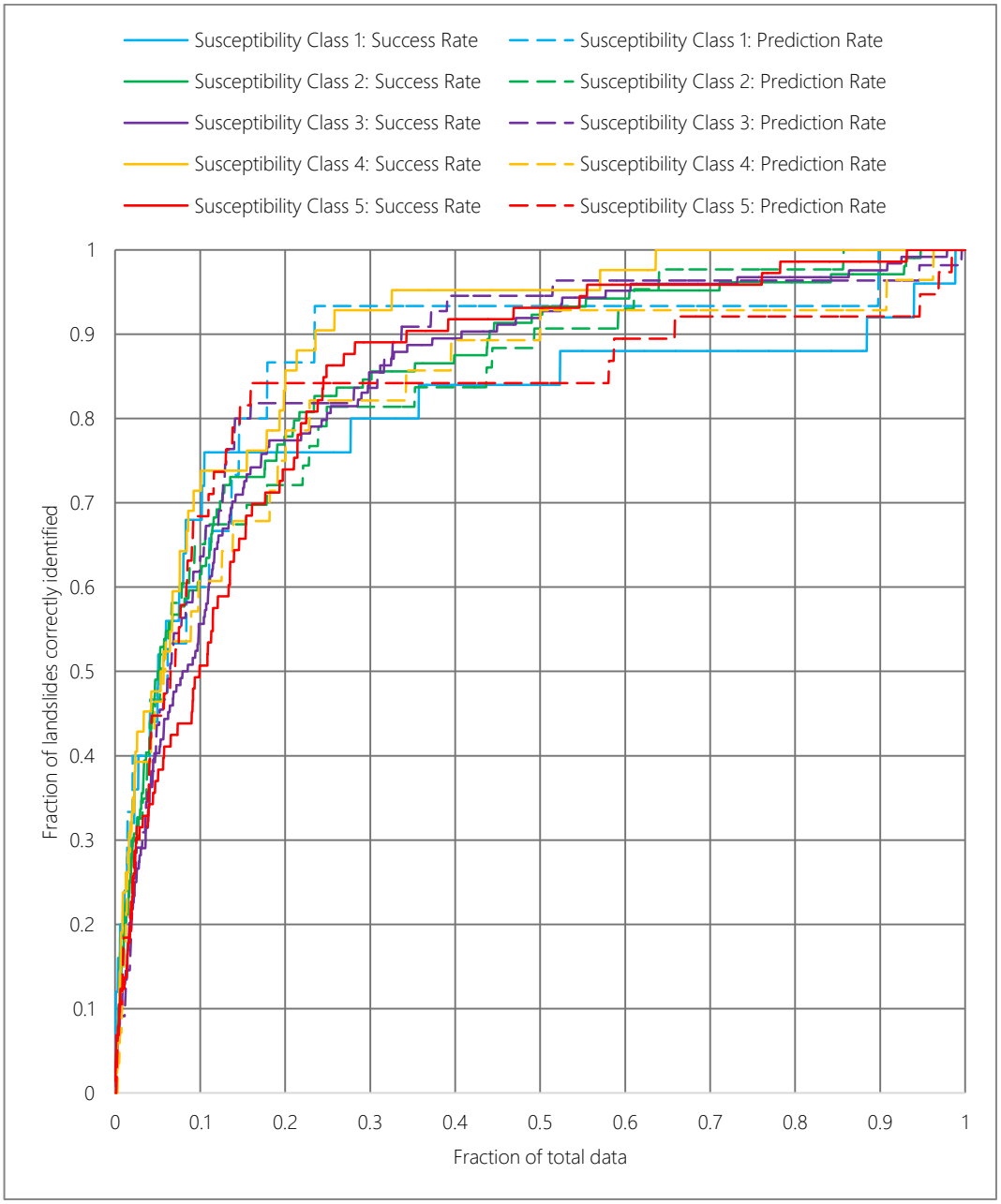


Figure 3 – ROC curve

3.2.5 Application of the Rainfall-Triggered Landslide Hazard Model at Global-Scale

Once the five models were calibrated, they were run for each day in the period 01/01/1980 – 31/12/2018. Results were then summed to generate 39 rainfall-triggered landslide hazard datasets – one for each year, each representing the modelled sum of landslides for that year. These annual summaries were the annualised for the period 1980 – 2018 using both the mean and the median average to produce *The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018)* and *The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)* (Section 5.4

includes more information on why two average rainfall-triggered landslide hazard maps have been produced).

3.2.6 Description of Rainfall-Triggered Landslide Hazard Assessment Results – Terminology

This approach uses the relationship between landslides reported as having occurred in populated, primarily English-speaking places and rainfall, to calibrate a series of models which estimate the probability of a landslide event occurring given any rainfall conditions anywhere in the world. Estimates of landslide probability (which is then annualised to give annual frequency) are therefore representative of *the sort of landslide that had it occurred in a populated, primarily English-speaking place, would have been reported*. This is clearly a clunky and un-useful, albeit technically precise description. It is therefore useful to look in further detail at the type of landslides reported in the NASA Global Landslide Catalogue. The catalogue contains some qualitative information on landslide size (Table 3).

Table 3 – Qualitative descriptions of the size distribution of events from the NASA Global Landslide Catalogue

Landslide size	Number of events	Percentage of total
Unknown	2,646	20%
Small	3,023	22%
Medium	6,889	51%
Large	866	6%
Very large	126	1%
Catastrophic	9	< 1%

If we assume those landslides for which the size was unknown must fall into one of the other five categories, the catalogue contains information on landslides which are either *small*, *medium*, *large*, *very large*, or *catastrophic*; or rather, landslides which are *small or larger*. The catalogue also contains records of landslides by numerous mechanisms, including debris flows, rock falls, translational landslides etc. It is however not possible to usefully compare a small rock fall to a small debris flow, nor a very large translational landslide to a medium-sized rotational landslide. Furthermore, it's not even really possible to compare one small debris flow with another small debris flow without some numerical benchmark against which *small* is defined.

It is therefore sensible at this stage of the analysis to take a step back from the quantitative logistic-regression-led analysis used to estimate the *annual frequency of the sort of landslide that had it occurred in a populated, primarily English-speaking place, would have been reported* and describe the results using more useful, qualitative and expert-judgement informed terminology. We propose the term *annual frequency of significant rainfall-triggered landslides* which accounts for the fact that the frequency estimates consider the size of landslides which are *small or larger*, are significant enough that would they have occurred in a

populated place they would have been notable, and that landslides can happen by any mechanism.

3.3 Earthquake-Triggered Landslide Hazard Assessment

3.3.1 General Approach

It is not possible to perform a similar logistic regression-based analysis for earthquake-triggered landslides as was done for rainfall-triggered landslides. This is primarily because the period of time for which both landslide and earthquake information is available is not of sufficient duration to represent a reasonable spatial distribution of long-term earthquake hazard. Numerical analysis by comparison of earthquake-triggered landslides from available inventories and the spatial distribution of earthquake triggers produces landslide hazard maps which are biased towards large, recent earthquakes.

A method based on that produced by Nadim et al. (2006; 2013) and later used by Arup-BGS (2018) is therefore used to estimate global earthquake-triggered landslide hazard.

Figure 4 presents an overview of the general approach for the estimation of quantitative earthquake-triggered landslide hazard.

3.3.2 Data Sources

In addition to the NASA Global Landslide Susceptibility Map (introduced in Section 2.3), the following data source was used for the earthquake-triggered landslide hazard assessment.

Global Seismic Hazard Assessment Program

The Global Seismic Hazard Assessment Program (GSHAP) (Giardini et al., 2003) was launched in 1992 by the International Lithosphere Program (ILP) with the support of the International Council of Scientific Unions (ICSU) and endorsed as a demonstration program in the framework of the United Nations International Decade for Natural Disaster Reduction (UN/IDNDR). The GSHAP project terminated in 1999, but still provides the best, freely available global seismic hazard mapping data for use by this project.



Figure 4 – Overview of the general approach for the estimation of quantitative earthquake-triggered landslide hazard

3.3.3 Pre-processing

Global Seismic Hazard Assessment Program

The Global Seismic Hazard Assessment Program data were pre-processed prior to use in the earthquake-triggered landslide hazard assessment.

- Data were cropped to the spatial extent of the NASA Global Landslide Susceptibility Map.
- Around the perimeter and at the northern and southern extremities of the NASA Global Landslide Susceptibility Map some seismic hazard values were missing. These were filled from the nearest adjacent areas.

3.3.4 Earthquake-Triggered Landslide Hazard Data analysis

Nadim et al. (2006; 2013) and Arup-BGS (2018) generate a landslide susceptibility score by scoring, ranking and multiplying a series of susceptibility factors. An earthquake-triggered landslide hazard score is then generated by multiplying this susceptibility score by a triggering factor. For this project, the susceptibility score is provided by the NASA Global Landslide Susceptibility Map – the score being a number between 1 and 5 representing very low through to very high landslide susceptibility. The earthquake-triggered landslide hazard score is generated by multiplying this NASA Global Landslide Susceptibility Map score by the seismic-trigger index proposed by Nadim et al. (2005; 2013) (Table 4). The resulting score is then scaled to the range 0 – 300 to allow comparison with the estimation of earthquake-triggered landslide hazard and the corresponding approximate annual frequency proposed by Nadim et al. (2006; 2013) (Table 5).

Table 4 – Seismic trigger index proposed by Nadim et al. (2006; 2013)

GSHAP PGA_{475} (m/s^2)	**Seismic trigger index**
0.0 – 0.5	0.1
0.5 – 1.0	0.4
1.0 – 1.5	0.8
1.5 – 2.0	1.5
2.0 – 2.5	2.5
2.5 – 3.0	3.5
3.0 – 3.5	5
3.5 – 4.0	6
4.0 – 4.5	7.5
> 4.5	10

Table 5 – Earthquake-triggered landslide hazard score and corresponding estimated annual frequency proposed by Nadim et al. (2006; 2013). For the two lowest scoring categories, Nadim et al. (2013) use a value of ~0 here, however to allow full calculation at all hazard levels we propose definite values.

Earthquake-triggered landslide hazard score	**Representative annual frequency / sq. km**
<7	0
8 – 24	0.00003
25 – 47	0.0001
48 – 74	0.0003
75 – 108	0.001
109 – 152	0.003
153 – 205	0.01
206 – 270	0.03
>270	0.1

3.3.5 Description of Earthquake-Triggered Landslide Hazard Assessment Results – Terminology

In line with the proposed terminology for rainfall-triggered landslides, the term annual frequency of significant earthquake-triggered landslides is proposed.

3.4 Assessment of Landslide Seasonality

Landslide seasonality describes the months throughout the year when rainfall-triggered landslide hazard is higher due to seasonal weather patterns.

The NASA global Landslide Hazard Assessment for Situational Awareness (LHASA) dataset has been used to assess landslide seasonality. High-Hazard Nowcasts are issued when sufficient rainfall occurs in an area of moderate or high landslide susceptibility (according to the NASA Global Landslide Susceptibility Map), as demonstrated in Figure 5.

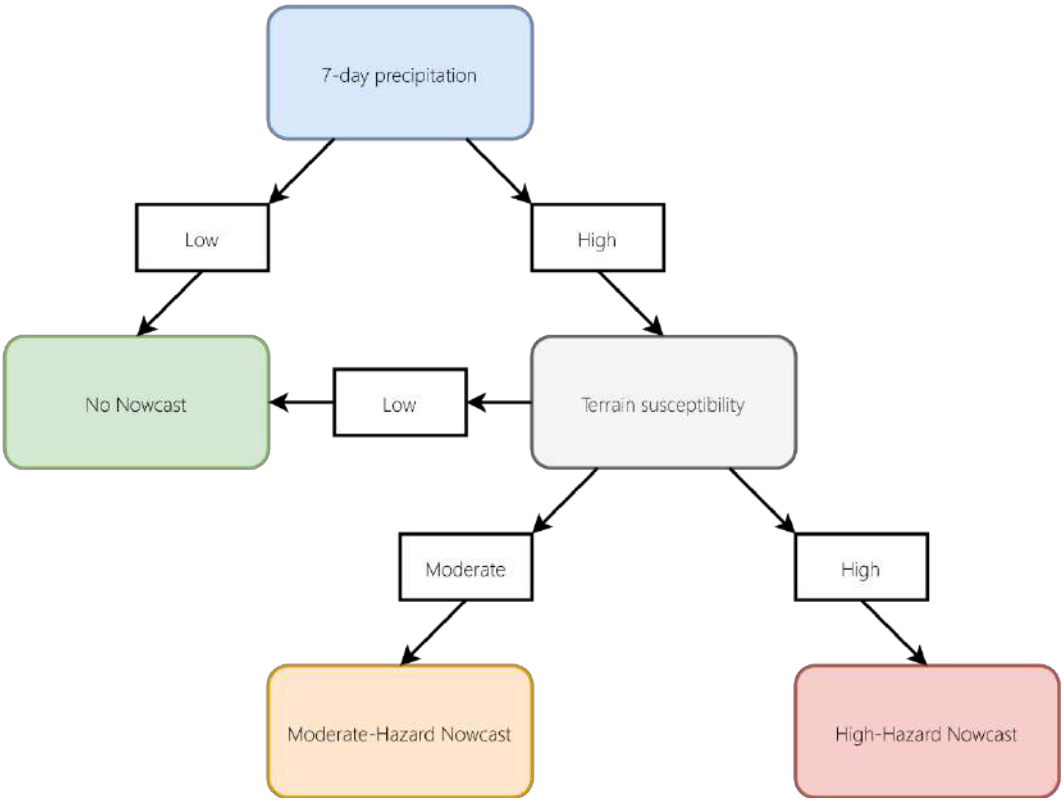


Figure 5 – Flowchart demonstrating the conditions required for a nowcast to be issued.

By counting the average number of issued nowcasts in a given month of the year for a particular area, it is possible to model which months of the year landslide hazard is highest. The results can be plotted on a histogram like the one shown in Figure 6.

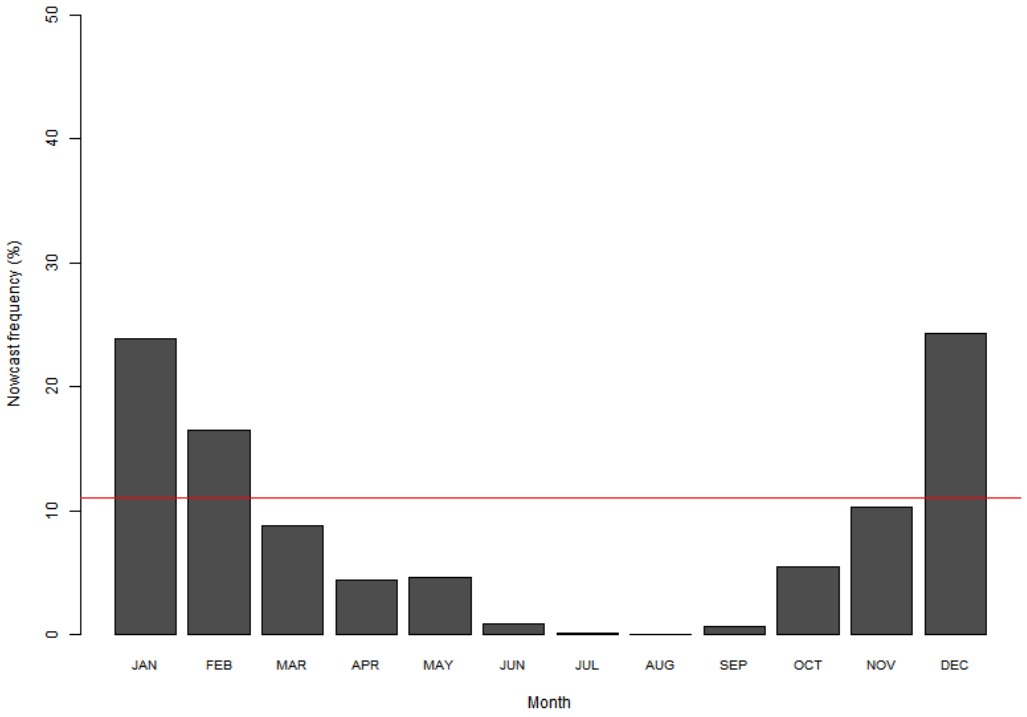


Figure 6 – Histogram showing the average frequency of moderate-hazard nowcasts for each month of the year for an example area. The threshold to delimit *seasonal* months is shown as a red line.

In order to describe in which months landslide hazard is highest, a threshold is required. It was decided that using quartiles was the best method for this threshold calculation and Equation 1 was adopted for defining *seasonal* months.

$$\text{Seasonal month nowcast frequency} \geq \text{Q3} + (0.5 \times \text{IQR})$$

Equation 1

Where Q3 is the third quartile and IQR is the interquartile range. Any months with a nowcast frequency greater than or equal to the seasonal month nowcast frequency are used as an indicator for months of high landslide hazard. In the example shown in Figure 6 the period of increased landside hazard described by landslide seasonality is during the months of December to February.

Landslide seasonality has been estimated at country scale and in countries for countries which are estimated to have high qualitative landslide hazard only.

3.5 Inclusion of Hazard Data in ThinkHazard!

3.5.1 Qualitative Descriptions of Hazard

For inclusion in ThinkHazard! the quantitative estimates of landslide hazard (average annual number of significant landslides / sq. km) must be converted into qualitative categorical descriptions of landslide hazard. Figure 7 presents an overview of the general approach for the estimation of qualitative landslide hazard.

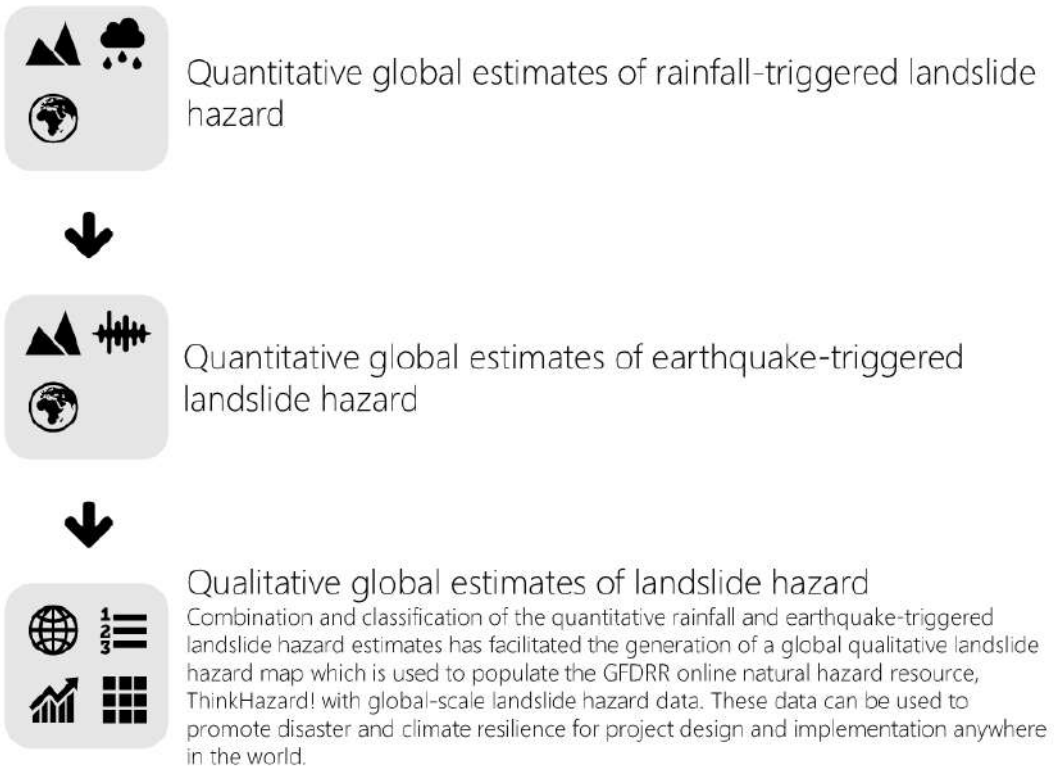


Figure 7 – Overview of the general approach for the estimation of qualitative landslide hazard

Nadim et al. (2013) use the classification shown in Table 6 (which is similar to the method currently used by ThinkHazard!).

ThinkHazard! requires further simplification to four hazard categories. For this project this is done by adding together the quantitative rainfall and earthquake-triggered landslide hazard datasets and using the classification shown in Table 7, which is based on the method of Nadim et al. (2013) and the existing methodology used by ThinkHazard!. Median estimated average annual rainfall-triggered landslide hazard values were used for this step to reduce the effect from event-driven hazard hotspots in the underlying data and to provide a more general overview of landslide hazard. These event-driven hazard hotspots are discussed further in Section 5.

Table 6 – Quantitative to qualitative classification of landslide hazard proposed by Nadim et al. (2012). For the two lowest qualitative categories (Negligible and Very low), Nadim et al. (2013) use a value of ~0 here, however to allow full calculation at all hazard levels we propose definite values.

Representative annual frequency / sq. km	Qualitative classification of landslide hazard
0	Negligible
0.00003	Very low
0.0001	Low
0.0003	Low to moderate
0.001	Moderate
0.003	Medium
0.01	Medium to high
0.03	High
0.1	Very high

Table 7 – Categorization of landslide hazard values for ThinkHazard!

ThinkHazard! hazard category	Proposed ranges of annual landslide frequency / sq. km
Very low	< 0.0001
Low	0.0001 - 0.001
Medium	0.001 - 0.01
High	> 0.01

3.5.2 Aggregation of Qualitative Data to Administration Units

It is necessary to aggregate qualitative hazard data from 1 km spatial resolution raster data to administrative units (polygons) for inclusion in ThinkHazard!. A new method for aggregating the qualitative hazard data from 1 km spatial resolution raster data to admin. level 2 level was established as part of this work. Further aggregation from admin. level 2 to admin. level 1 and then from admin. level 1 to admin. level 0 was done using the existing ThinkHazard! methodology, whereby the larger admin. level is characterised by the maximum value from the admin. units which it contains.

Several methods were developed, trialled and the results discussed at a project workshop. It was decided that the 80th percentile was the most suitable method of aggregation. The 80th percentile method gave a good visual fit with the disaggregated raster data, capturing the importance of high hazard value pixels for the aggregated classification. This means that in a defined administrative unit:

- The aggregated value assigned is the value above which only 20% of the pixel values may be found.

This also means that if in a defined administrative unit:

- At least 80% of qualitative landslide hazard raster pixels are either Very low, Low, Medium or High, the aggregated qualitative landslide hazard score would be High;

- At least 80% of qualitative landslide hazard raster pixels are either Very low, Low or Medium, the aggregated qualitative landslide hazard score would be Medium;
- At least 80% of qualitative landslide hazard raster pixels are either Very low or Low, the aggregated qualitative landslide hazard score would be Low; or
- At least 80% of qualitative landslide hazard raster pixels are Very low, the aggregated qualitative landslide hazard score would be Very low.

An example of the 80th percentile aggregation for Ethiopia is shown on Figure 8 and Figure 9.

Any admin. units that fall entirely outside extent of the hazard raster (approximately 60°S to 72°N) were not aggregated. Any admin. units that fall partially outside of the hazard raster were classified according to the part of the admin. unit that falls within the hazard raster.

3.5.3 Preparation of Accompanying Text

The landslide hazard is defined as the estimated average annual frequency of occurrence of a significant landslide occurring within a defined area. The probability of a given size event or larger to occur is an exceedance probability. The average time interval between each event being exceeded is therefore inverse of the exceedance probability or the recurrence interval.

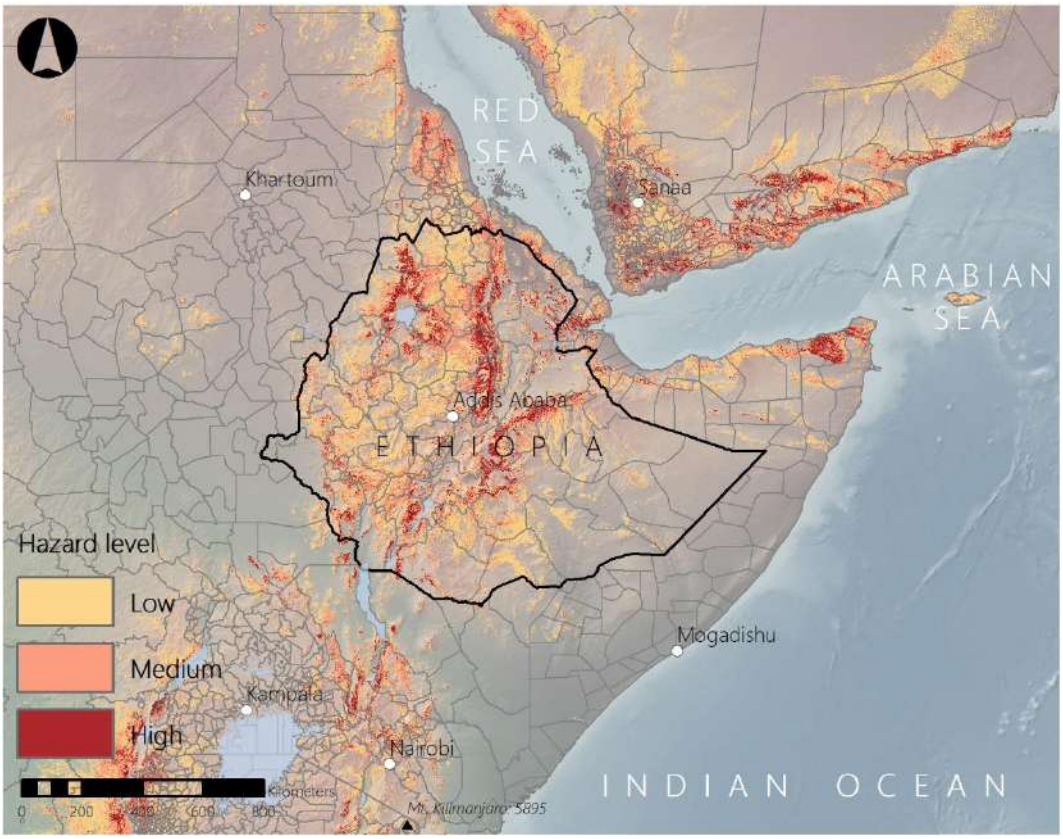


Figure 8 – Disaggregated landslide hazard estimates from *The Global Landslide Hazard Map: Qualitative Landslide Hazard*

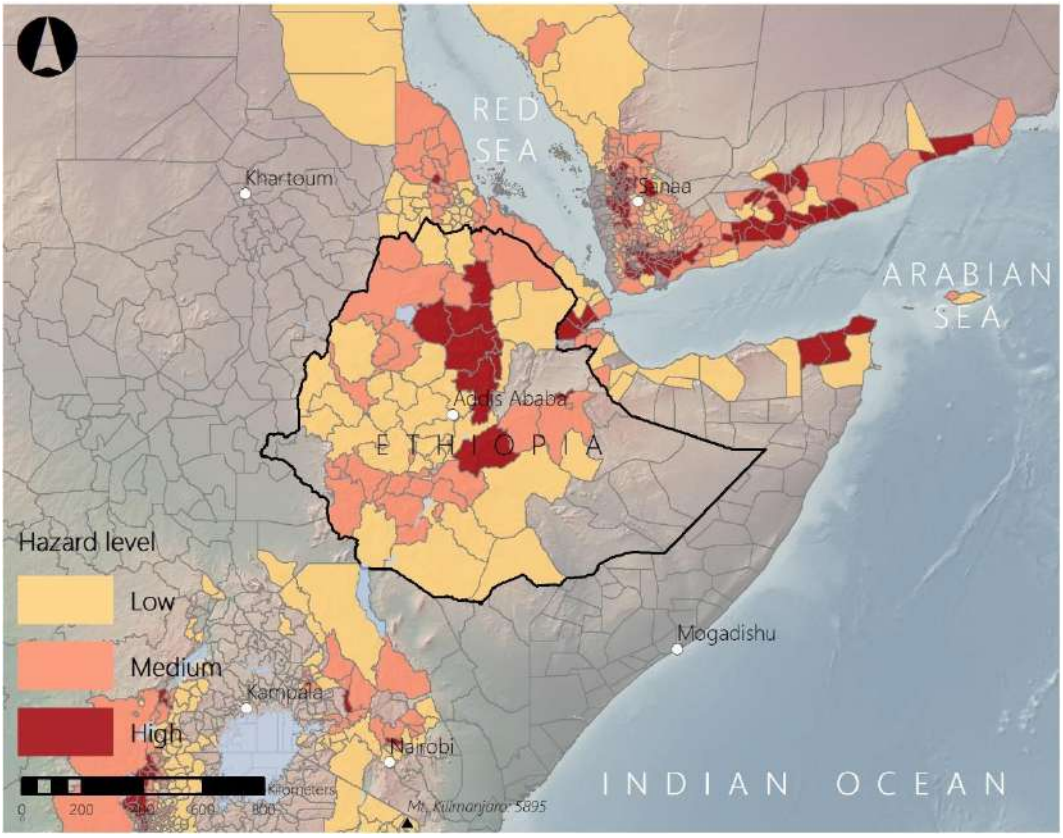


Figure 9 – Admin. level 2 aggregated landslide hazard estimates from *The Global Landslide Hazard Map: Admin. Level 2 Qualitative Landslide Hazard*

Recurrence interval terminology sometimes tends to be more understandable by practitioners, e.g. terms like a recurrence interval of 10 years or a 1 in 10-year event, appear familiar. However, it is important to avoid the misconception of regularity of time between events. Statistical methods have been used and it is therefore correct to assume that the events are independent in time (i.e. that the probability distribution of events is not changing through time). In simple terms this means that two events of the same average annual frequency or recurrence interval could by chance occur at a shorter time interval but on average over a long period of time the average annual frequency or recurrence intervals should be correct.

The text to be presented in ThinkHazard! alongside the interactive landslide hazard map is contained in the sections below. Certain words in this text are modified according to the landslide hazard level of each admin. unit. Modified words are shown in **bold**.

It was agreed that seasonality is only reported for countries that also have a high landslide hazard. The seasonality text is as follows:

"Landslide hazard is highest during the month(s) of **list of month(s)**."

Very low hazard

"In the area you have selected, landslide hazard is classified as **very low** according to the information that is currently available. This means that a significant landslide is likely to occur **less than once every 10,000 years per square kilometer (equivalent to an annual frequency of <0.0001 per square kilometer).** Based on this information, planning decisions such as project siting, project design, and construction methods, **may** take into account the potential for landslides. Further detailed information should be obtained to better understand the level of landslide susceptibility in your project area."

Low hazard

"In the area you have selected, landslide hazard is classified as **low** according to the information that is currently available. This means that a significant landslide is likely to occur **less than once every 1,000 and more than once every 10,000 years per square kilometer (equivalent to an annual frequency of 0.0001-0.001 per square kilometer).** Based on this information, planning decisions such as project siting, project design, and construction methods, **may** take into account the potential for landslides. Further detailed information should be obtained to better understand the level of landslide susceptibility in your project area."

Medium hazard

"In the area you have selected, landslide hazard is classified as **medium** according to the information that is currently available. This means that a significant landslide is likely to occur **less than once every 100 years and more than once every 1,000 years per square kilometer (equivalent to an annual frequency of 0.001-0.01 per square kilometer).** Based on this information, planning decisions such as project siting, project design, and construction methods, **should** take into account the potential for landslides.

Further detailed information should be obtained to better understand the level of landslide susceptibility in your project area."

High hazard

"In the area you have selected, landslide hazard is classified as **high** according to the information that is currently available. This means that a significant landslide is likely to occur **more than once every 100 years per square kilometer (equivalent to an annual frequency of >0.01 per square kilometer).** Based on this information, planning decisions such as project siting, project design, and construction methods, **must** take into account the potential for landslides. Further detailed information should be obtained to better understand the level of landslide susceptibility in your project area."

4 Global Landslide Hazard Results

4.1 Introduction

This section of the report provides a summary of the key findings from the global landslide hazard assessment. The results are presented in the form of a series of global maps and tabulated results at national and global scale.

4.2 Global Overview of Landslide Hazard

- It is estimated that in the order of 15,500,000 significant rainfall-triggered landslides occurred worldwide between 1980 and 2018.
- The average annual number of significant rainfall-triggered landslides to occur globally is estimated to be in the order of 400,000 based on data from 1980 – 2018.
- The average annual number of significant earthquake-triggered landslides is estimated to be in the order of 130,000.

Table 8 presents the top ten highest rainfall-triggered landslide hazard countries by estimated average annual number of significant landslides and Table 9 presents the top ten highest rainfall-triggered landslide hazard countries by estimated average annual number of significant landslides per square kilometer.

Appendix B contains full tabulated estimates of landslide hazard by country.

Appendix C contains full tabulated estimates of landslide seasonality by country.

Table 8 – Estimated average annual number of significant rainfall-triggered landslides (1980-2018)

Country	Country code	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)
United States of America	259	36,150
China	147295	35,280
India	115	31,430
Philippines	196	23,110
Indonesia	116	22,220
Russian Federation	204	18,340
Myanmar	171	15,080
Brazil	37	13,360
Canada	46	11,780
Vietnam	264	11,490

Table 9 – Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km (excluding countries with an area < 10,000 sq. km)

Country	Country code	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Taiwan	147296	0.2006
Comoros	58	0.1915
Réunion	206	0.1183
Philippines	196	0.0781
Bangladesh	23	0.0435
French Polynesia (Fr.)	87	0.0419
Disputed Area[5]	15	0.0411
Vietnam	264	0.0351
Vanuatu	262	0.0351
Malaysia	153	0.0329

Table 10 presents the top ten highest earthquake-triggered landslide hazard countries by estimated average annual number of significant landslides and Table 11 presents the top ten highest earthquake-triggered landslide hazard countries by estimated average annual number of significant landslides per square kilometre.

Table 10 – Estimated average annual number of significant earthquake-triggered landslides

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides
China	147295	20,950
Kyrgyz Republic	138	11,070
United States of America	259	10,710
Turkey	249	9,270
Mexico	162	7,510
Islamic Republic of Iran	117	6,580
Russian Federation	204	6,290
Tajikistan	239	5,810
Afghanistan	1	4,990
Nepal	175	4,550

[5] The eastern-most disputed area on the border between India and China to the east of Bhutan.

Table 11 – Estimated average annual number of significant earthquake-triggered landslides / sq. km (excluding countries with an area < 10,000 sq. km)

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides / sq. km
Taiwan	147296	0.0811
Kyrgyz Republic	138	0.0556
Tajikistan	239	0.0409
Disputed Area[5]	15	0.0337
Costa Rica	61	0.0317
Nepal	175	0.0309
Vanuatu	262	0.022
New Zealand	179	0.0168
Solomon Islands	225	0.0164
Turkey	249	0.0119

4.3 Rainfall-Triggered Landslide Hazard (1980-2018)

Figure 10 shows the estimated average annual number of significant rainfall-triggered landslides between 1980 and 2018.

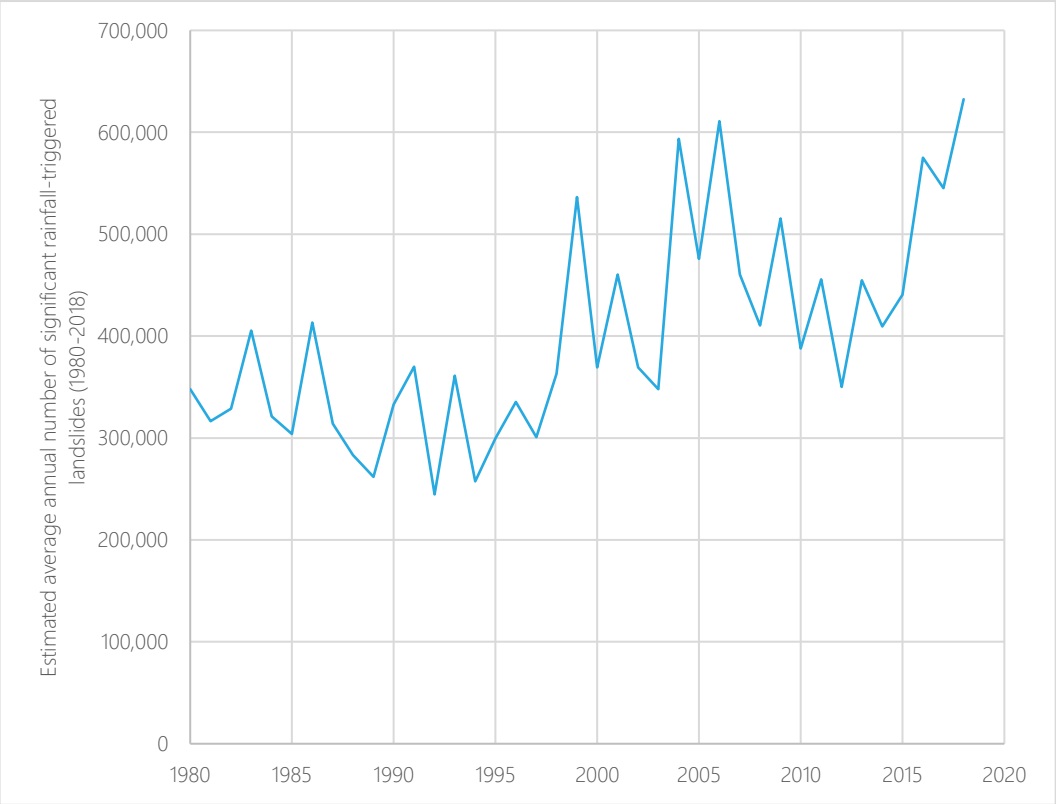


Figure 10 – Estimated average annual number of significant rainfall-triggered landslides worldwide (1980–2018)

A trend over the 38-year period of increasing estimated average annual numbers of rainfall-triggered landslides is interpreted to reflect increasing frequency and intensity of rainfall events. The average year-on-year increase is equal to approximately 1 – 2%. It is possible this reflects a broad climate change effect at global scale.

Figure 11 shows the estimated average annual number of significant rainfall-triggered landslides between 1980 and 2018 sub-divided by broad geographic regions[6].

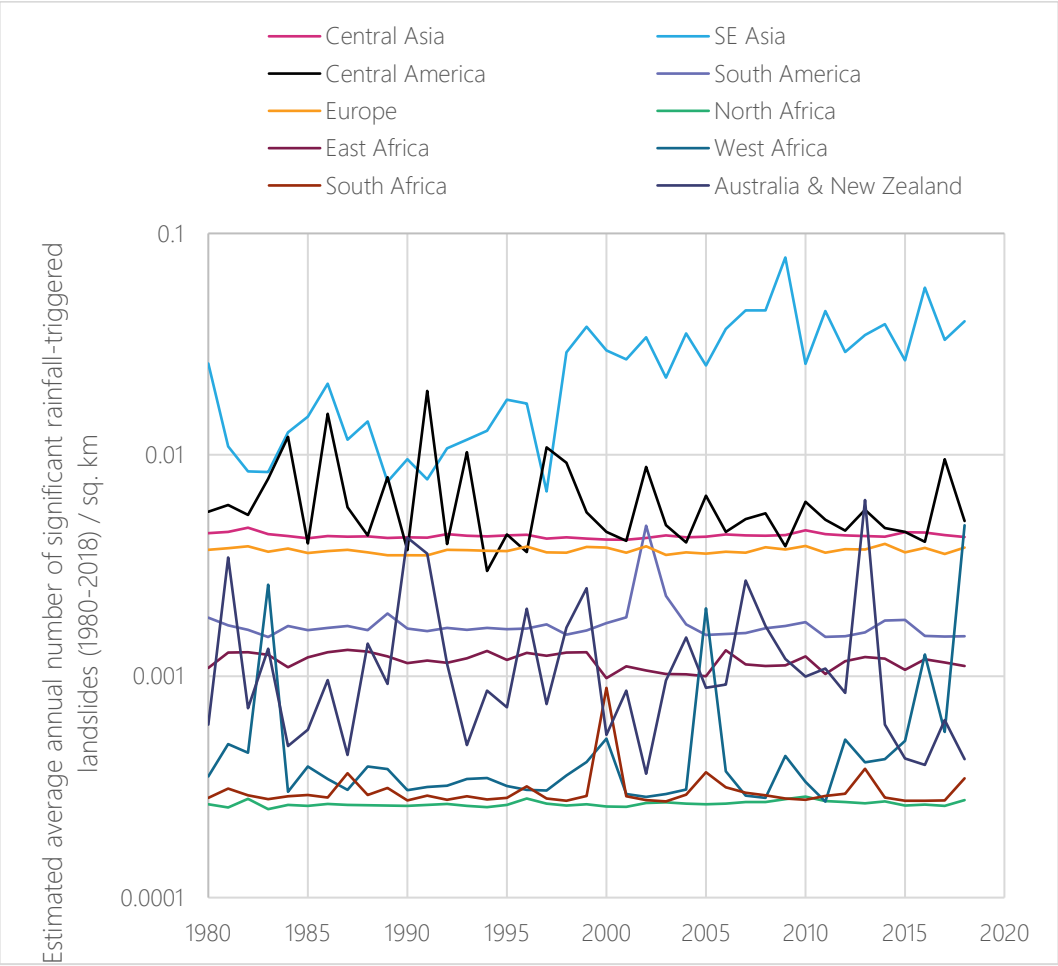


Figure 11 – Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km sub-divided into broad geographic regions[6]

[6] Sample countries for each broad geographic region: Central Asia: Uzbekistan, Tajikistan, Kyrgyz Republic, Afghanistan and Pakistan. South East Asia: Taiwan, Philippines, Malaysia, Indonesia, Vietnam and Cambodia. Central America: Mexico, Guatemala, Honduras, El Salvador, Nicaragua, Costa Rica and Panama. South America: Chile, Argentina, Uruguay, Paraguay and Bolivia. Europe: United Kingdom, France, Germany, Spain, Italy, Portugal and Switzerland. North Africa: Morocco, Algeria and Libya. East Africa: Ethiopia, Uganda, Kenya and South Sudan. West Africa: Sierra Leone, Guinea, Ghana and Nigeria. South Africa: South Africa, Lesotho, Namibia, Botswana and Zimbabwe.

Figure 10 shows a broad upward trend in the average annual number of significant rainfall-triggered landslides estimated to have occurred between 1980 and 2018. As shown by Figure 11, this trend is driven by high and increasing levels of landslide hazard in South East Asia. Figure 11 also shows that landslide hazard is much more consistent and less event-driven on the whole than in areas which are not (or less) affected by tropical, monsoonal rainfall.

Trends shown by Figure 10 and Figure 11 show the impact of rainfall on landslide hazard isolated from the underlying change with time of landslide susceptibility caused by anthropogenic action (landslide susceptibility and the contribution to this by anthropogenic action is based on a ~2018 baseline). It is therefore plausible that in the early parts of the time-series included in this analysis (at which time global population was much less than in 2018), the contribution to landslide susceptibility made by anthropogenic action is overestimated and hence so too are resulting estimates of landslide hazard. This could mean that the average year-on-year increase in rainfall-triggered landslide hazard estimated to be approximately 1 – 2% (1980-2018) may be an underestimate.

4.4 Global Landslide Hazard Maps

This project has produced five new global-scale landslide hazard maps:

The Global Landslide Hazard Map: Qualitative Landslide Hazard

The Global Landslide Hazard Map: Qualitative Landslide Hazard presents a combined representation of global landslide hazard at a global scale. It is the combination of the sum of *The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)* and *The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard* which has then been simplified to four categories, ranging from Very low to High landslide hazard, based on the existing system used by ThinkHazard!.

The Global Landslide Hazard Map: Qualitative Landslide Hazard is included in Appendix A.

The Global Landslide Hazard Map: Admin. Level 2 Qualitative Landslide Hazard

The Global Landslide Hazard Map: Admin. Level 2 Qualitative Landslide Hazard shows the qualitative landslide hazard levels aggregated to admin. level 2 at a global-scale, presented as they are shown in ThinkHazard!.

The Global Landslide Hazard Map: Admin. Level 2 Qualitative Landslide Hazard is included in Appendix A.

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018)

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018) presents an estimate of rainfall-triggered landslide hazard which is representative for the period 1980-2018. The mean average landslide

hazard values are in some parts of the world very clearly influenced by major rainfall/storm events which have occurred between 1980 and 2018.

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018) is included in Appendix A.

The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)

The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018) presents an estimate of rainfall-triggered landslide hazard which is representative for the period 1980-2018 but is less influenced by incidences of increased landslide hazard due to major rainfall/storm events which have occurred between 1980 and 2018. This representation of landslide hazard is more generalised than *The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018)*.

The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018) is included in Appendix A.

The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard

The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard presents an estimate of earthquake-triggered landslide hazard at a global scale which builds upon work by Nadim et al. (2006; 2013) and uses the NASA Global Landslide Susceptibility Map as a base.

The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard is included in Appendix A.

5 Discussion of Global Landslide Hazard Results

5.1 Introduction

This section of the report provides a discussion of some of the key findings from the project and provides comparison with the results of other studies at a global and regional scale.

5.2 Comparison with Other Landslide Inventories

It should be noted that there are limited landslide inventories available for validation of results. Due to the global-scale of this project, only selected landslide inventories have been used for validation at this stage.

Arizona Natural Hazards Viewer

The Arizona Natural Hazards Viewer (Arizona Geological Survey, 2019) contains information on ~6,000 landslides in Arizona, USA. In one area in southeast Arizona there are a series of 173 mapped debris flows, each recorded as *Historic, < 150 years [old]* over an area of ~250 sq.km. Assuming this is a reasonably complete record for this area, this corresponds to annual frequency of landslides of [173 landslides / 150 years / 250 sq. km] 0.0046 /sq. km. This project estimates the mean average annual frequency of significant rainfall-triggered landslides in this area to be broadly 0.0001 – 0.01 /sq. km (Figure 12).

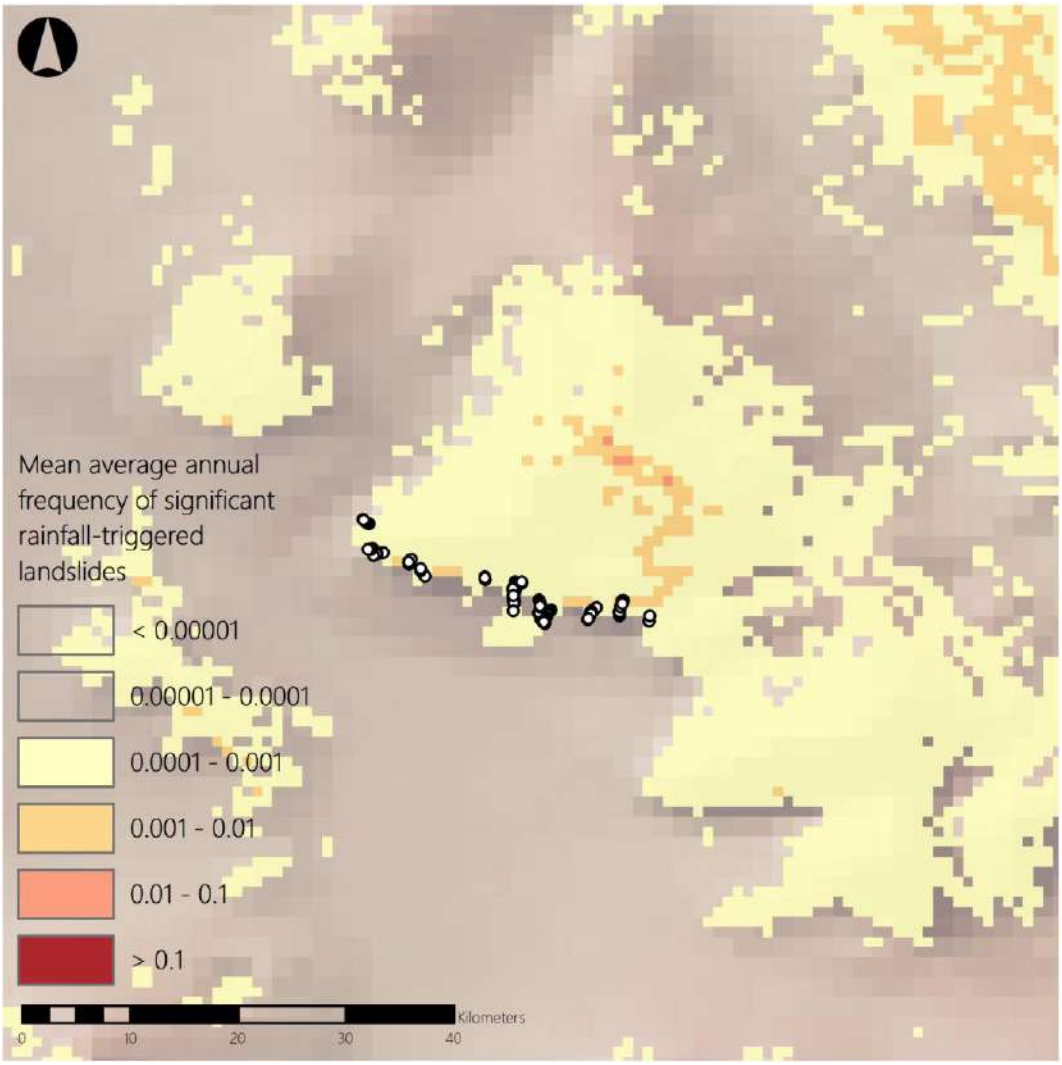


Figure 12 –Spatial correlation between *The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard* and an extract of a landslide inventory from the Arizona Natural Hazards Viewer. White points show locations of mapped landslides.

Msilimba & Holmes (2010) – Malawi

In part of a study by Msilimba & Holmes (2010) in Malawi, several landslides were mapped in the Chiweta area of Rumphi District, northern Malawi. These landslides were all mapped in areas identified by *The Global Landslide Hazard Map: Qualitative Landslide Hazard* as being higher hazard than the surrounding area (Figure 13).

Claessens et al. (2007) – Uganda

Claessens et al. (2007) compiled an inventory of 81 landslides around Mt. Elgon, Bududa District, Uganda. Although few inferences can be made about the frequency of landslide activity here, the spatial distribution is well matched to that of *The Global Landslide Hazard Map: Qualitative Landslide Hazard* (Figure 14).

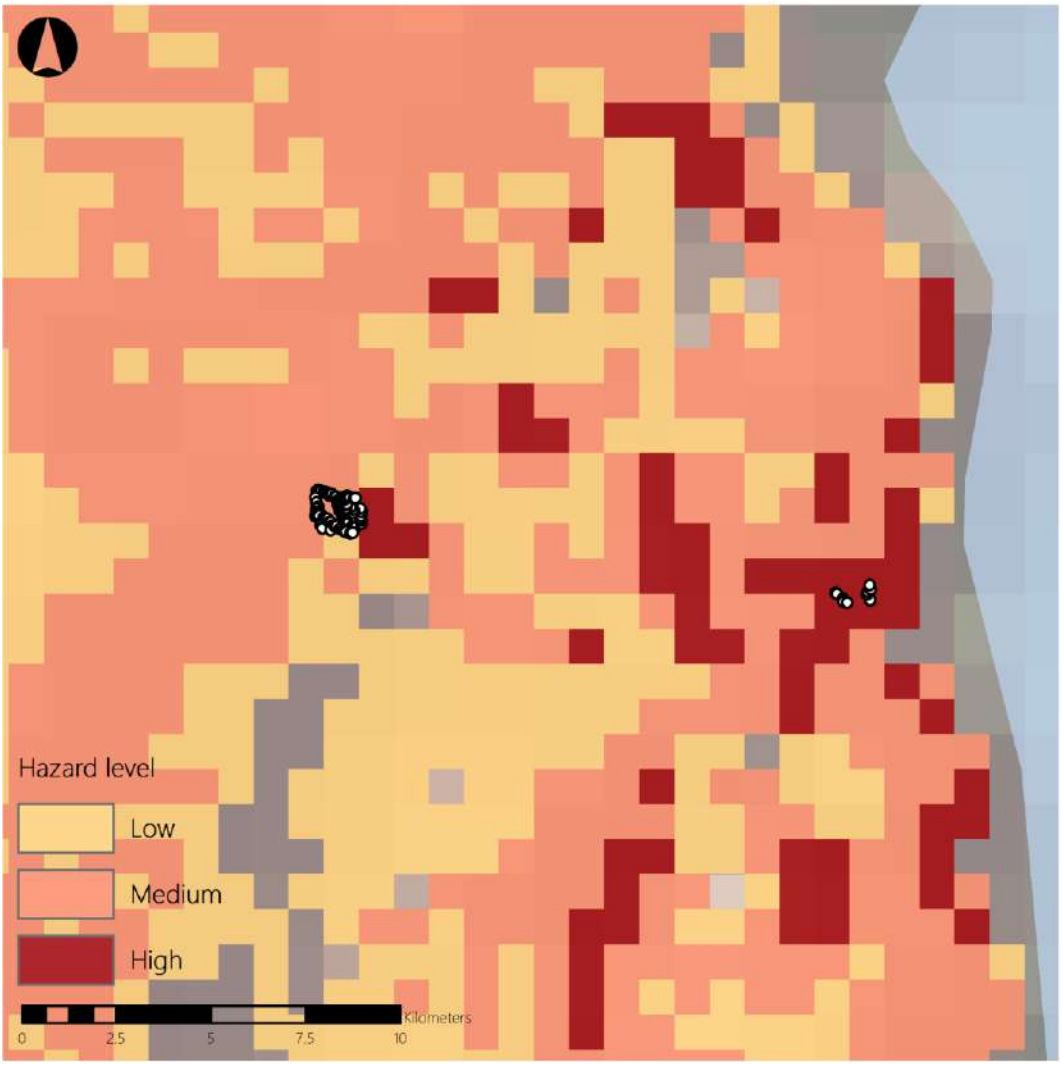


Figure 13 –Spatial correlation between landslides mapped by Msilimba & Holmes (2010) and *The Global Landslide Hazard Map: Qualitative Landslide Hazard* on the western shore of Lake Malawi, Malawi. White points show locations of mapped landslides.

Hong Kong ENTLI

Landslide hazard in Hong Kong is well understood and provides a useful case-study by which to benchmark the results of this study.

- The mean average annual number of significant rainfall-triggered landslides estimated by this study for Hong Kong is 20.
- The mean average annual number of significant rainfall-triggered landslides / sq. km estimated by this study for Hong Kong is 0.0168.
- Malone and Shelton (1982) recorded 97 landslides between 1978 and 1980, equivalent to an annual average of ~32 landslides. The Hong Kong Enhanced Natural Terrain Landslide Inventory (ENTLI) (GEO, 2019) recorded 4,911 landslides which were identified using aerial photography and have a minimum width of 10m between 1924 and 2013, equivalent to an annual average of ~55 landslides.

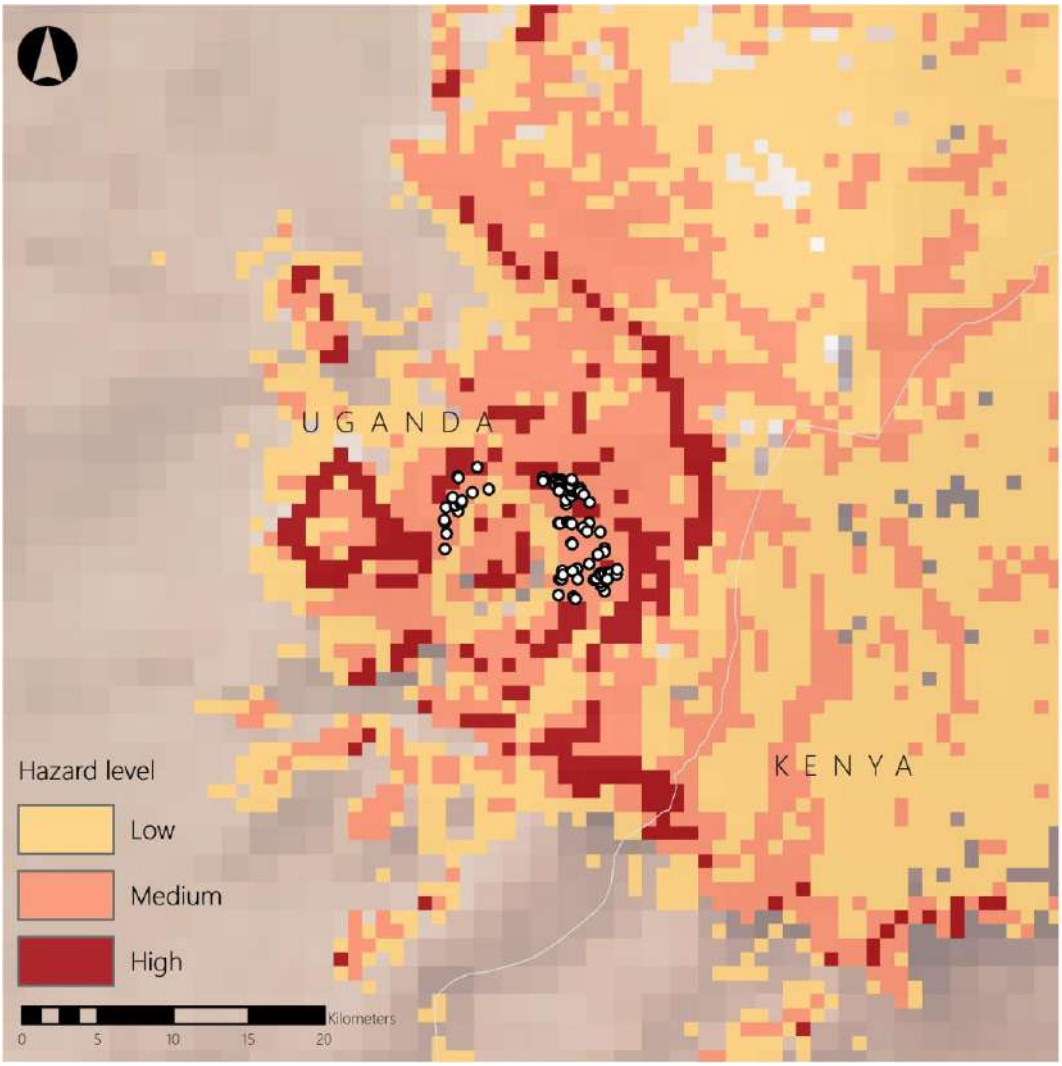


Figure 14 –Spatial correlation between the landslide inventory of Claessens et al. (2007) and *The Global Landslide Hazard Map: Qualitative Landslide Hazard* around Mt. Elgon, Bududa District, Uganda. White points show locations of mapped landslides.

5.3 Comparison with Previous Studies

Arup-BGS (2018) – National-Level Landslide Risk Profiles for Sub-Saharan Africa

Table 12 summarises the landslide hazard results for nine Sub-Saharan African countries as estimated by this study and Table 13 summarises the corresponding estimates from Arup-BGS (2018).

Table 12 – Landslide hazard in nine Sub-Saharan African countries as estimated by this study.

Country	Estimated average annual number of significant earthquake-triggered landslides	Estimated mean average annual number of significant rainfall-triggered landslides
Ethiopia	20	2,400
Kenya	<10	440
Uganda	<10	130
Niger	<10	40
Senegal	<10	<10
Malawi	<10	100
Mali	<10	20
Mozambique	<10	240
Cape Verde	<10	<10

Table 13 – Landslide hazard in nine Sub-Saharan African countries as estimated by Arup-BGS (2018)

Country	Estimated average annual number of earthquake-triggered landslides >1,000 – 3,000 sq. m (lower estimate)	Estimated average annual number of rainfall-triggered landslides >1,000 – 3,000 sq. m (lower estimate)
Ethiopia	1,490	2,220
Kenya	310	430
Uganda	130	190
Niger	NA	590
Senegal	NA	80
Malawi	80	120
Mali	NA	670
Mozambique	220	420
Cape Verde	10	20

Comparison of these hazard estimates broadly indicates that the mean average annual frequency of rainfall-triggered landslides estimated by this study is similar to the lower estimate by Arup-BGS (2018) for landslides which have an area of 1,000 – 3,000 sq. m or larger. Additionally, the average annual frequency of earthquake-triggered landslides estimated by this study is less than the frequency of earthquake-triggered landslides estimated by Arup-BGS (2018).

Estimates of the average annual number of significant earthquake-triggered landslides for this project were derived from GSHAP. One of the well documented limitations of GSHAP is the underestimation of seismic hazard in certain parts of the world, including in parts of Africa. Estimates of the average annual number of earthquake-triggered landslides by Arup-BGS (2018) are based on project-specific estimates of seismic hazard developed as part of the *R5* project (introduced in Section 1.1). These project-specific seismic hazard studies provide a much higher resolution assessment of seismic hazard in these nine countries. This difference in study-scale and approach may account for the difference in estimates for earthquake-triggered landslide hazard for the nine countries in Sub-Saharan Africa.

The different periodicity of earthquake and rainfall landslide triggers makes benchmarking of annual estimates challenging. Whilst average annual rainfall-

triggered landslide frequency might be reasonably evaluated over a relatively short period (say ~10 years or more), to get a full appreciation of earthquake-triggered landslide hazard a much longer period of observation is required. This is due to the potential for very large but infrequent events (both in terms of earthquake magnitude and resulting earthquake-triggered landslides). Marc et al. (2019) (summarised by Kornei, 2019) estimate the contribution to landslide-related erosion from earthquake-triggered landslides to be between 10 and 50 – 60% (assuming the remainder are due to rainfall-triggering).

Arup et al. (2019) – Sierra Leone Multi-City Hazard Review and Risk Assessment

As part of a World Bank-funded project to understand, map and quantify multi-hazard risk for three cities in Sierra Leone, Arup, BGS, JBA and INTEGEMS produced a quantitative landslide hazard assessment for the city of Freetown, Sierra Leone. The project was an exhaustive city-scale landslide hazard assessment based on a local landslide inventory (Figure 15) and supported by fieldwork validation.

The quantitative landslide hazard assessment undertaken as part of the Sierra Leone Multi-City Hazard Review and Risk Assessment estimated the average annual number of landslides >100 sq. m to affect the Freetown Peninsula to be 7 – 74. This project estimates the mean average annual number of significant rainfall-triggered landslides to affect the same area to be ~10.

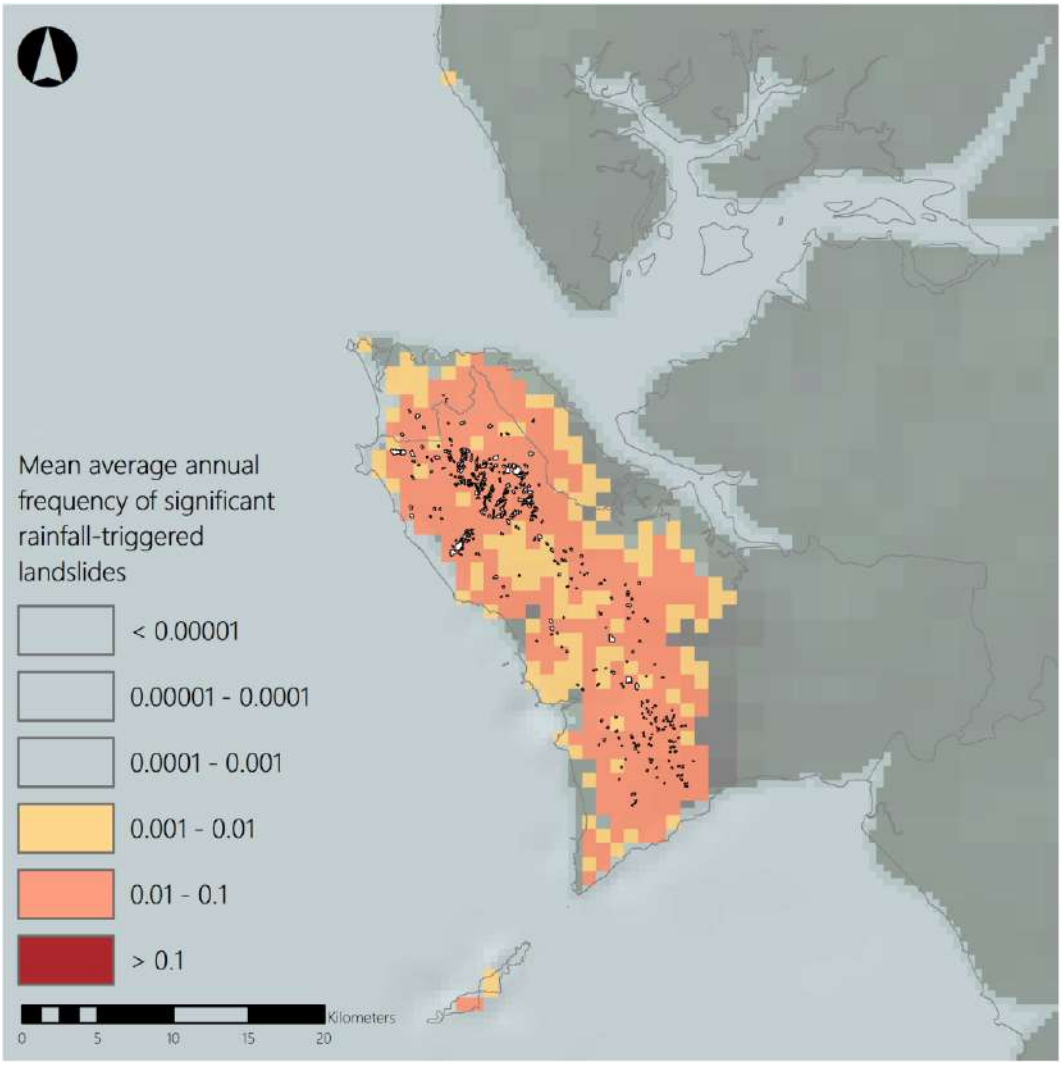


Figure 15 –Spatial correlation between the landslide inventory of Arup et al. (2019) and *The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard* for Freetown, Sierra Leone. White polygons show locations of mapped landslides.

Existing data on ThinkHazard!

It is possible to perform a large-scale comparison between the spatial coverage achieved by this study and the existing coverage offered by ThinkHazard!. Figure 16 and Figure 17 present a comparison for Ethiopia. These comparisons show both a significant improvement in general coverage and identify areas of high landslide hazard which were previously not included on ThinkHazard!.

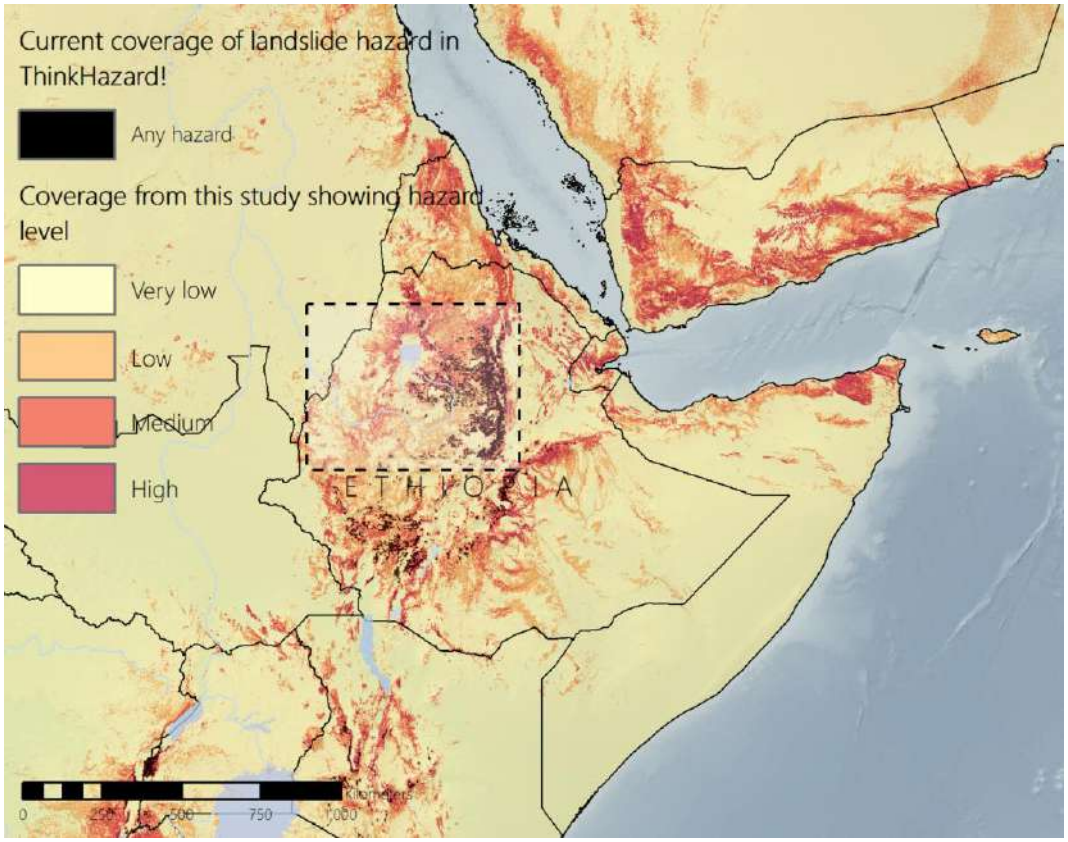


Figure 16 – Comparison of the spatial coverage and hazard levels of this study with the existing data on ThinkHazard! for Ethiopia. Dashed box shows extent of Figure 17.

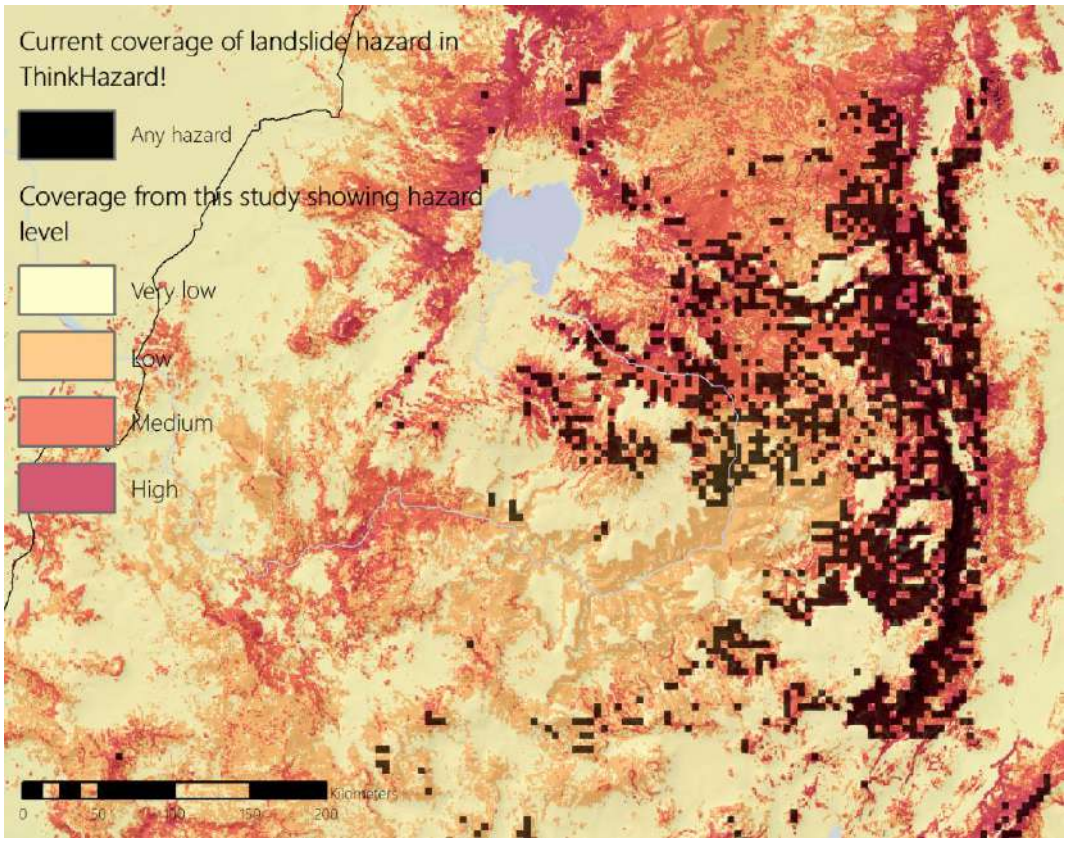


Figure 17 – Comparison of the spatial coverage and hazard levels of this study with the existing data on ThinkHazard! for central Ethiopia. Extent shown as dashed box on Figure 16.

5.4 Effect of Large Storm Events

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018) contains some areas of increased landslide hazard due to major rainfall/storm events which have occurred between 1980 and 2018. For example, increased rainfall associated with Hurricane Mitch, which affected Central America in 1998 (and indeed caused numerous landslides – Harp et al., Unknown Year) resulted in the estimated average annual frequency of significant rainfall-triggered landslides (1998) to be elevated above the average for this region (Figure 18). This increase in hazard is such that the mean estimated average annual frequency of significant rainfall-triggered landslides (1980-2018) is elevated in the same area (Figure 19).

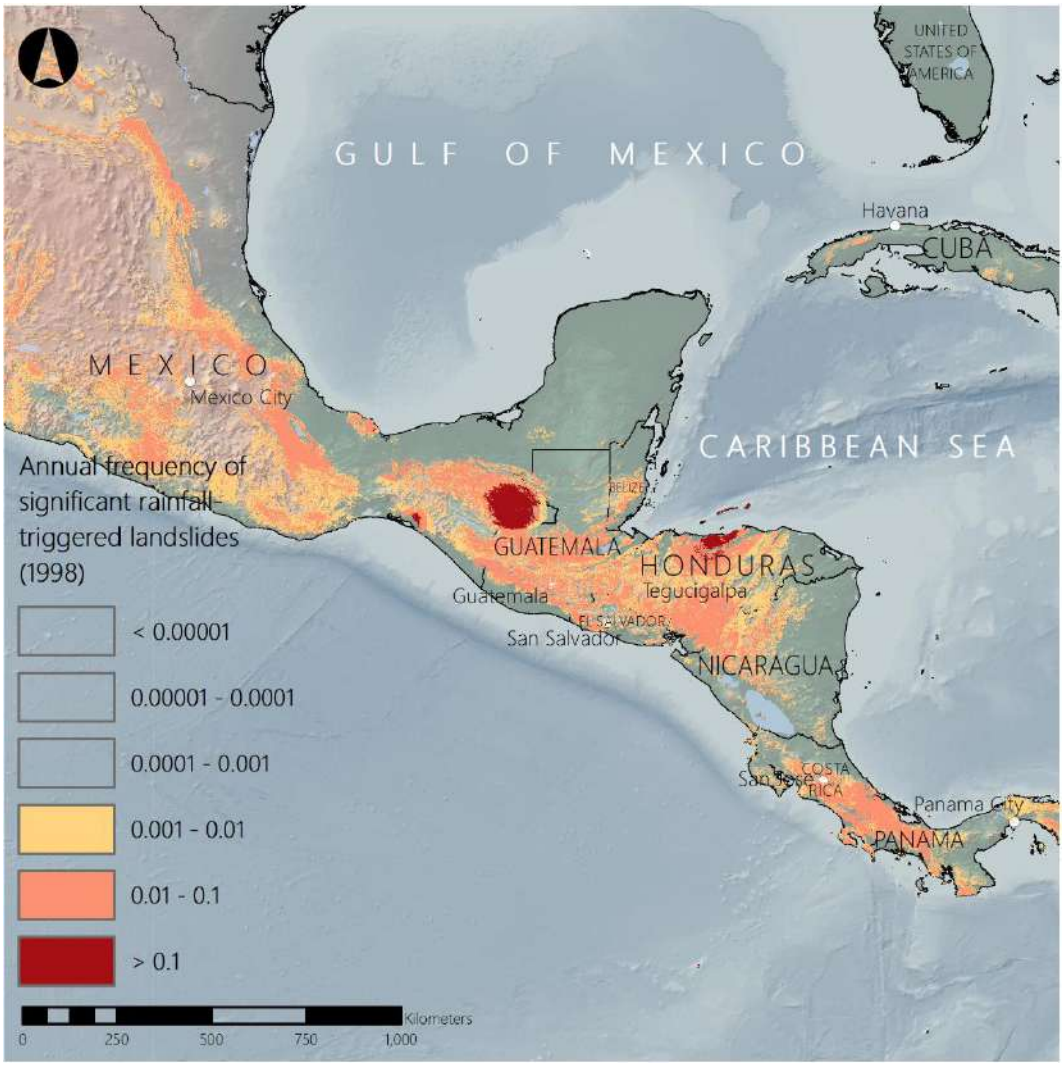


Figure 18 – Annual frequency of significant rainfall-triggered landslides (1998) around the Gulf of Mexico. Elevated due to increased rainfall associated with Hurricane Mitch.

There are numerous other examples of the *footprint* left by increased areas of landslide hazard associated with storm events in *The Global Landslide Hazard*

Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018), many of which are around the Gulf of Mexico, but there are also some isolated examples in South America and Eastern-Central Africa.

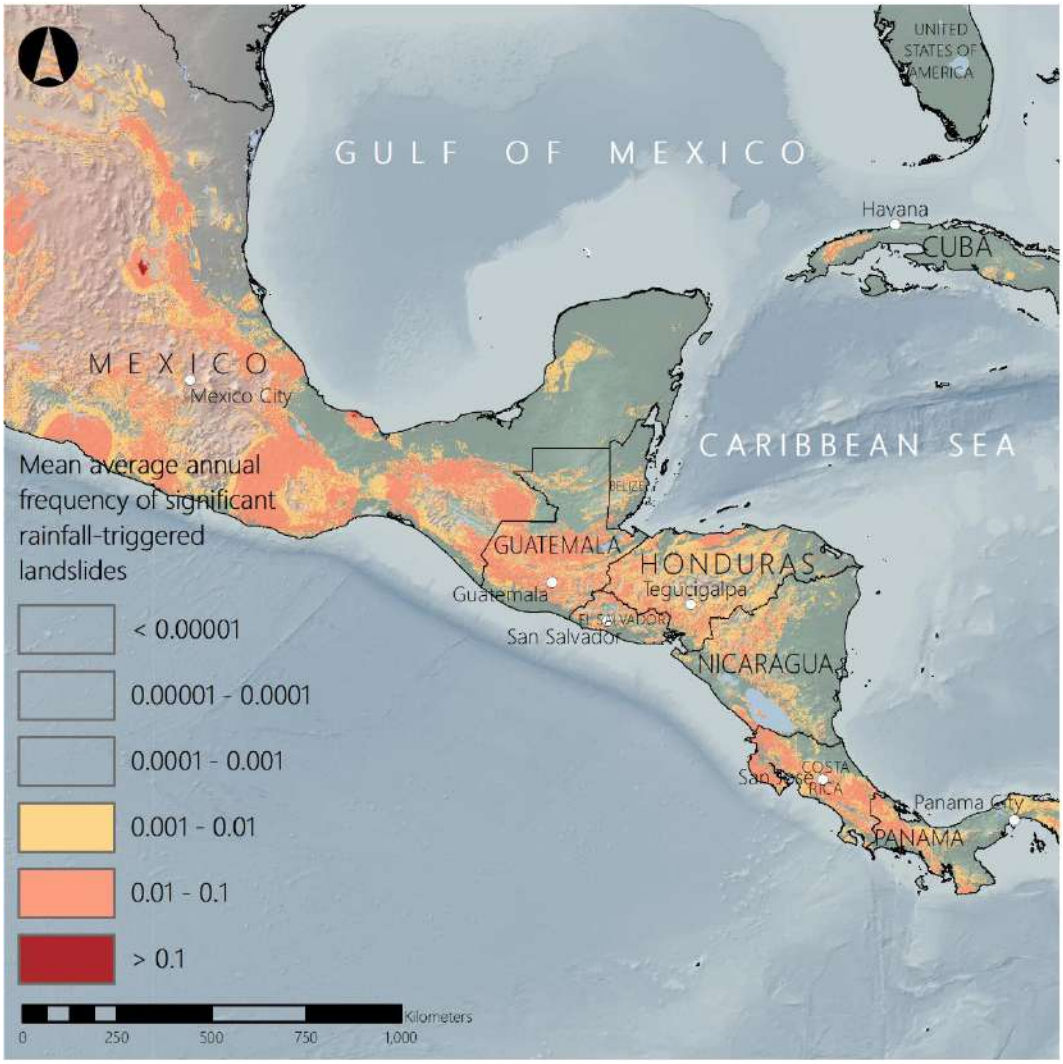


Figure 19 – Mean estimated average annual frequency of significant rainfall-triggered landslides (1980-2018) around the Gulf of Mexico. Shows areas of increased landslide hazard which can be directly associated with Hurricane Mitch.

There are numerous other examples of the *footprint* left by increased areas of landslide hazard associated with storm events in *The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018)*, many of which are around the Gulf of Mexico, but there are also some isolated examples in South America and Eastern-Central Africa.

It is for this reason that both mean and median-based rainfall-triggered landslide hazard maps are presented. This is also why the median-version of the map was used for definition of qualitative landslide hazard rather than the mean.

5.5 Significant Landslide Size

A significant landslide, as defined in Section 3.2.6, is *the sort of landslide that had it occurred in a populated, primarily English-speaking place, would have been reported*. It is therefore not straightforward to characterise a significant landslide by a minimum size (again, as per Section 3.2.6, we note that significant landslides must be of a certain minimum size or larger and cannot simply represent any landslide of any size). It is however helpful, at least for better interpreting the statistics associated with global landslide hazard, to assign a nominal minimal size to a significant landslide. As discussed in Section 5.3, the mean average annual frequency of significant rainfall-triggered landslides estimated by this study is similar to the lower estimate by Arup-BGS (2018) for landslides which have an area of 1,000 – 3,000 sq. m or larger. This would suggest a minimum landslide size of say 1,000 – 3,000 sq. m. Likewise, the mean average annual frequency of significant rainfall-triggered landslides estimated by this project for Freetown, Sierra Leone, was lower than the estimate by Arup et al. (2019) where the minimum modelled landslide size was ~100 sq. m. This would indicate a minimum size for significant landslides in excess of 100 sq. m (which is perhaps unsurprising given the difference in resolution between the two studies).

On the basis of the limited information available we propose that a significant landslide has a minimum size of 1,000 sq. m. We further note that given the uncertainty associated with this estimate, the simply by referring to results using the term *significant landslide* as discussed is likely to be more widely understandable and arguably useful. Figure 20 – Figure 24 present a selection of examples of what might constitute a *significant landslide* of ~1,000 sq. m in area



Figure 20 – Example of what might constitute a *significant landslide* (Sierra Leone, 2019, photo credit: UNOPS)



Figure 21 – Example of what might constitute a *significant landslide* (China, 2014, photo credit: Peter Redshaw)



Figure 22 – Example of what might constitute a *significant landslide* (Isle of Wight, 2012, photo credit: Peter Redshaw)



Figure 23 – Example of what might constitute a *significant landslide* (Malawi, 2016, photo credit: Ngwira, 2016)



Figure 24 – Example of what might constitute a *significant landslide* (Cabo Verde, 2009, photo credit: ASemana, 2009)

5.6 Error handling

Despite pre-processing and thorough QA checking, given the broad temporal and spatial resolution of the datasets being used, some erroneous results were identified.

Central America, Early 90's Rainfall Data

Review of the rainfall-triggered landslide hazard data identified an unusual peak in global rainfall-triggered landslide hazard in 1991 and 1992. This was traced back to results in Mexico, where annual landslide hazard estimates were 50 – 100x higher than for previous years. Further examination of the underlying NOAA CPC Rainfall Data identified anomalous data in both 1991 and 1992 which directly accounted for these anomalous landslide hazard values. Rainfall-triggered landslide hazard estimates for the corresponding areas of Mexico for 1991 and 1992 were omitted from the results presented herein.

Artefacts in global-scale datasets

The production of global-scale maps using global-scale data brings with it a wealth of challenges and inevitably means that some of the datasets used to generate global maps will be imperfect – perhaps patchy in places or missing data. Some such artefacts from the NASA Global Landslide Susceptibility Map and the

Global Seismic Hazard Assessment Program Map are carried through the calculations and are evident in the global landslide hazard maps.

6 Summary and Conclusions

6.1 Introduction

This section of the report provides high level summary and conclusions. The overall aim of the project was to develop a global-scale landslide hazard assessment. This landslide hazard assessment is intended to be used to generate improved global coverage for landslide hazard levels on ThinkHazard!.

6.2 Summary of Landslide Hazard Results

This project has produced systematically determined, quantitative estimates of rainfall-triggered landslide hazard and earthquake-triggered landslide hazard on a 1 km grid for any location in the world. The quantitative estimates of rainfall and earthquake-triggered landslide hazard have been combined to produce a global qualitative landslide hazard map for inclusion in ThinkHazard!.

It is estimated that a total in the order of 15,500,000 significant rainfall-triggered landslides occurred worldwide between 1980 and 2018. The average annual number of significant rainfall-triggered landslides to occur globally is estimated to be in the order of 400,000 based on the analysis of data from 1980 to 2018. The average annual number of significant earthquake-triggered landslides is estimated to be in the order of 130,000.

A global trend of increasing rainfall-triggered landslide hazard is identified which is equivalent to an average year-on-year increase of 1-2% between 1980 and 2018. It is possible this reflects a broad climate change effect at global scale.

Full tabulated estimates of landslide hazard by country are provided in Appendix B.

6.2.1 Landslide Hazard Maps

Five new global-scale landslide hazard maps are presented in this report:

The Global Landslide Hazard Map: Qualitative Landslide Hazard

The Global Landslide Hazard Map: Qualitative Landslide Hazard presents a combined representation of global landslide hazard at a global scale. It is the combination of the sum of *The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)* and *The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard* which has then been simplified to four categories, ranging from Very low to High landslide hazard, based on the existing system used by ThinkHazard!. This map is also presented aggregated to admin. level 2 as **The Global Landslide Hazard Map: Admin. Level 2 Qualitative Landslide Hazard**.

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018)

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018) presents an estimate of rainfall-triggered landslide hazard which is representative for the period 1980-2018. The mean average landslide hazard values are in some parts of the world very clearly influenced by major rainfall/storm events which have occurred between 1980 and 2018. A variant of this map, **The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)** is also presented. The median average results appear to be less influenced by incidences of increased landslide hazard due to major rainfall/storm events which have occurred between 1980 and 2018.

The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard

The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard presents an estimate of earthquake-triggered landslide hazard at a global scale which builds upon work by Nadim et al. (2006; 2013) and uses the NASA Global Landslide Susceptibility Map as a base.

6.2.2 Landslide Seasonality

An assessment of landslide seasonality using a statistical analysis of the NASA Global Landslide Hazard Assessment for Situational Awareness (LHASA) has been undertaken. Landslide seasonality describes the months throughout the year when rainfall-triggered landslide hazard is highest due to seasonal weather patterns. The analysis shows distinct potential landslide seasonality trends for different geographies. The seasonality results for all countries are summarised in Appendix C.

6.3 What is new about *The Global Landslide Hazard Map* in comparison to existing maps?

- This project has developed the first global inventory-based quantitative estimates of rainfall-triggered landslide hazard. These estimates have been shown to be broadly in-line with previous expert judgement-driven global studies.
- Where data from *The Global Landslide Hazard Map* are brought into ThinkHazard! they provide hazard information in areas where previously no hazard information was calculated.
- Using rainfall data from between 1980 and 2018 this project has developed a 39-year estimate of *real* landslide hazard – i.e. one which is based upon the spatial and temporal distribution of actual rainfall recorded between 1980 and 2018 (subject to the limitations of accuracy of the rainfall data).
- The estimates of both rainfall and earthquake-triggered landslide hazard are based on the NASA Global Landslide Susceptibility Map, which provides a

consistent landslide susceptibility input across the globe, including up-to-date anthropogenic as well as environmental susceptibility factors.

7 Recommendations for Future Work

7.1 Introduction

This section of the report provides recommendations for future work. The recommendations are provided for ways to improve and enhance the landslide hazard assessment and for ways in which the landslide hazard assessment results can be taken forward to provide an improved understanding of landslide risk at a global scale.

7.2 Recommendations

Climate Change

It would be of great benefit to apply climate change projections to the models developed by this project to estimate future levels of rainfall-triggered landslide hazard. This is particularly the case since this project identifies broadly increasing levels of global rainfall-triggered landslide hazard between 1980 and 2018.

Rainfall Data

The NOAA CPC Rainfall Data provided a suitable source of readily available rainfall data with the required spatial and temporal resolution to support this project. However, it may be possible to source improved rainfall data to facilitate refined estimates of landslide hazard.

Global Landslide Risk

By overlaying the data generated by this project onto existing models of population and the built environment it would be possible to extend the estimates of hazard through to estimates of risk at a global scale. This could provide new insights into global landslide hazard which can, for the first time, be robustly described in a quantitative way.

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Appendix A

Global Landslide Hazard Maps

A1 Summary of Global Landslide Hazard Maps

The Global Landslide Hazard Map: Qualitative Landslide Hazard

The Global Landslide Hazard Map: Qualitative Landslide Hazard presents a combined representation of global landslide hazard at a global scale. It is the combination of the sum of *The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)* and *The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard* which has then been simplified to four categories, ranging from Very low to High landslide hazard, based on the existing system used by ThinkHazard!.

The Global Landslide Hazard Map: Admin. Level 2 Qualitative Landslide Hazard

The Global Landslide Hazard Map: Admin. Level 2 Qualitative Landslide Hazard shows the qualitative landslide hazard levels aggregated to admin. level 2 at a global-scale, presented as they are shown in ThinkHazard!.

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018)

The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018) presents an estimate of rainfall-triggered landslide hazard which is representative for the period 1980-2018. The mean average landslide hazard values are in some parts of the world very clearly influenced by major rainfall/storm events which have occurred between 1980 and 2018.

The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)

The Global Landslide Hazard Map: Median Annual Rainfall-Triggered Landslide Hazard (1980-2018) presents an estimate of rainfall-triggered landslide hazard which is representative for the period 1980-2018 but is less influenced by incidences of increased landslide hazard due to major rainfall/storm events which have occurred between 1980 and 2018. This representation of landslide hazard is more generalised than *The Global Landslide Hazard Map: Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018)*.

The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard

The Global Landslide Hazard Map: Earthquake-Triggered Landslide Hazard presents an estimate of earthquake-triggered landslide hazard at a global scale which builds upon work by Nadim et al. (2006; 2013) and uses the NASA Global Landslide Susceptibility Map as a base.

Appendix B

Tabulated Global Landslide Hazard Estimates

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Abyei	102	<10	<10	<0.0001	<0.0001
Afghanistan	1	4990	3190	0.0078	0.005
Albania	3	30	510	0.0009	0.0179
Algeria	4	10	590	<0.0001	0.0003
American Samoa (U.S.)	5	<10	<10	<0.0001	0.0003
Andorra	7	<10	<10	<0.0001	0.007
Angola	8	<10	140	<0.0001	0.0001
Anguilla (U.K.)	9	<10	<10	<0.0001	0.0029
Antigua and Barbuda	11	<10	<10	0.0004	0.0002
Arab Republic of Egypt	40765	<10	340	<0.0001	0.0003
Argentina	12	260	2440	0.0001	0.0009
Armenia	13	80	190	0.0028	0.0065
Aruba (Neth.)	14	<10	<10	<0.0001	0.0001
Ashmore and Cartier Islands (Aus.)	16	N/A	N/A	N/A	N/A
Australia	17	30	8630	<0.0001	0.0011
Austria	18	<10	1310	<0.0001	0.0156
Azerbaijan	19	300	410	0.0018	0.0025
Azores Islands (Por.)	74578	<10	20	<0.0001	0.0065

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Bahrain	21	<10	<10	<0.0001	<0.0001
Baker Island (U.S.)	22	<10	<10	<0.0001	<0.0001
Bangladesh	23	30	6090	0.0002	0.0435
Barbados	24	<10	<10	<0.0001	0.0015
Belarus	26	<10	<10	<0.0001	<0.0001
Belgium	27	<10	30	<0.0001	0.0011
Belize	28	<10	70	<0.0001	0.003
Benin	29	<10	10	<0.0001	0.0001
Bermuda (U.K.)	30	<10	<10	<0.0001	<0.0001
Bhutan	31	110	640	0.0029	0.017
Bolivia	33	110	3110	0.0001	0.0029
Bonaire (Neth.)	75001	<10	<10	<0.0001	0.0001
Bosnia and Herzegovina	34	20	430	0.0004	0.0083
Botswana	35	<10	<10	<0.0001	<0.0001
Bouvet Island (Nor.)	36	<10	<10	<0.0001	0.0052
Brazil	37	10	13360	<0.0001	0.0016
British Indian Ocean Territory (U.K.)	38	<10	<10	<0.0001	0.0032
British Virgin Islands (U.K.)	39	<10	<10	0.0001	0.0142

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Brunei Darussalam	40	<10	30	<0.0001	0.0056
Bulgaria	41	30	210	0.0003	0.0019
Burkina Faso	42	<10	<10	<0.0001	<0.0001
Burundi	43	<10	100	<0.0001	0.0038
Cambodia	44	<10	3890	<0.0001	0.0215
Cameroon	45	<10	1760	<0.0001	0.0038
Canada	46	360	11780	<0.0001	0.0012
Cape Verde	47	<10	<10	<0.0001	0.0007
Cayman Islands (U.K.)	48	<10	<10	0.0001	0.0001
Central African Republic	49	<10	50	<0.0001	0.0001
Chad	50	<10	140	<0.0001	0.0001
Chile	51	1900	3480	0.0025	0.0046
China	147295	20950	35280	0.0022	0.0038
Christmas Island (Aus.)	54	<10	<10	<0.0001	0.0004
Clipperton Island (Fr.)	55	<10	<10	<0.0001	<0.0001
Cocos (Keeling) Islands (Aus.)	56	<10	<10	<0.0001	<0.0001
Colombia	57	910	7200	0.0008	0.0063
Comoros	58	<10	320	<0.0001	0.1915

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Congo	59	<10	160	<0.0001	0.0005
Cook Islands (N.Z.)	60	<10	<10	<0.0001	0.0046
Costa Rica	61	1620	790	0.0317	0.0154
Côte d'Ivoire	66	<10	70	<0.0001	0.0002
Croatia	62	20	300	0.0004	0.0054
Cuba	63	20	360	0.0002	0.0032
Curaçao (Neth.)	75002	<10	<10	<0.0001	0.0001
Cyprus	64	<10	30	0.0005	0.0038
Czech Republic	65	<10	140	<0.0001	0.0017
D. P. R. of Korea	67	<10	470	<0.0001	0.0038
Democratic Republic of Congo	68	10	1010	<0.0001	0.0004
Denmark	69	<10	<10	<0.0001	<0.0001
Disputed Area[7]	52	30	20	0.0081	0.0052
Disputed Area[8]	2	40	150	0.0012	0.0048
Disputed Area[9]	15	2270	2770	0.0337	0.0411
Djibouti	70	<10	100	0.0001	0.0046

[7] Disputed area on the border between India and China.
[8] The northern-most disputed area on the border between India and China to the east of Pakistan.
[9] The eastern-most disputed area on the border between India and China to the east of Bhutan.

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Dominica	71	<10	<10	0.0002	0.0028
Dominican Republic	72	40	220	0.0009	0.0046
Ecuador	73	710	2830	0.0028	0.0111
El Salvador	75	40	180	0.0018	0.0086
Equatorial Guinea	76	<10	190	<0.0001	0.0072
Eritrea	77	<10	210	<0.0001	0.0017
Estonia	78	<10	<10	<0.0001	<0.0001
Ethiopia	79	20	2400	<0.0001	0.0021
Falkland Islands (Malvinas)	81	<10	<10	0.0001	0.0003
Faroe Islands (Den.)	82	<10	<10	<0.0001	0.0006
Federated States of Micronesia	163	<10	<10	0.0001	0.0054
Fiji	83	<10	490	<0.0001	0.0266
Finland	84	<10	30	<0.0001	0.0001
France	85	10	1720	<0.0001	0.0031
French Guiana	86	<10	30	<0.0001	0.0003
French Polynesia (Fr.)	87	<10	170	<0.0001	0.0419
French Southern and Antarctic Lands (Fr.)	88	<10	<10	<0.0001	0.0005
FYR of Macedonia	241	10	60	0.0003	0.0023

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Gabon	89	<10	1250	<0.0001	0.0047
Georgia	92	460	1000	0.0066	0.0144
Germany	93	<10	660	<0.0001	0.0018
Ghana	94	<10	50	<0.0001	0.0002
Gibraltar (U.K.)	95	<10	<10	<0.0001	0.0193
Greece	97	220	1000	0.0016	0.0075
Greenland (Den.)	98	<10	220	<0.0001	0.0001
Grenada	99	<10	<10	0.0001	0.0008
Guadeloupe	100	<10	10	0.0003	0.0051
Guam (U.S.)	101	<10	10	0.0001	0.0147
Guatemala	103	190	1300	0.0017	0.0119
Guernsey (U.K.)	104	<10	<10	<0.0001	0.0017
Guinea	106	<10	290	<0.0001	0.0012
Guinea-Bissau	105	<10	<10	<0.0001	<0.0001
Guyana	107	<10	110	<0.0001	0.0005
Haiti	108	<10	280	0.0001	0.0103
Heard Island and McDonald Islands (Aus.)	109	<10	<10	<0.0001	0.0111
Honduras	111	30	910	0.0003	0.0081

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Hong Kong, SAR	33364	<10	20	<0.0001	0.0168
Howland Island (U.S.)	112	<10	<10	<0.0001	<0.0001
Hungary	113	<10	40	<0.0001	0.0004
Iceland	114	150	190	0.0015	0.0018
India	115	3600	31430	0.0012	0.0102
Indonesia	116	690	22220	0.0004	0.0118
Iraq	118	280	330	0.0007	0.0008
Ireland	119	<10	50	<0.0001	0.0008
Islamic Republic of Iran	117	6580	5540	0.0039	0.0033
Isle of Man (U.K.)	120	<10	<10	<0.0001	0.0004
Israel	121	<10	40	0.0002	0.0022
Italy	122	70	2530	0.0002	0.0084
Jamaica	123	10	150	0.0007	0.0135
Japan	126	2000	7090	0.0053	0.019
Jarvis Island (U.S.)	127	<10	<10	<0.0001	<0.0001
Jersey (U.K.)	128	<10	<10	<0.0001	0.0004
Johnston Atoll (U.S.)	129	<10	<10	<0.0001	<0.0001
Jordan	130	<10	70	<0.0001	0.0008

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Kazakhstan	132	3090	760	0.0011	0.0003
Kenya	133	<10	440	<0.0001	0.0008
Kingman Reef (U.S.)	134	N/A	N/A	N/A	N/A
Kiribati	135	<10	<10	<0.0001	<0.0001
Kosovo	75008	<10	30	0.0001	0.0031
Kuril islands	136	<10	<10	0.0001	0.0003
Kuwait	137	<10	<10	<0.0001	<0.0001
Kyrgyz Republic	138	11070	1820	0.0556	0.0091
Lao People's Democratic Republic	139	10	4180	0.0001	0.0182
Latvia	140	<10	<10	<0.0001	<0.0001
Lebanon	141	20	100	0.0015	0.0099
Lesotho	142	<10	20	<0.0001	0.0007
Liberia	144	<10	1560	<0.0001	0.0162
Libya	145	<10	90	<0.0001	0.0001
Liechtenstein	146	<10	<10	<0.0001	0.028
Lithuania	147	<10	<10	<0.0001	<0.0001
Luxembourg	148	<10	10	<0.0001	0.0038
Macau, SAR	149	<10	<10	<0.0001	0.0085

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Madagascar	150	<10	5550	<0.0001	0.0094
Madeira Islands (Por.)	151	<10	<10	<0.0001	0.0003
Malawi	152	<10	100	<0.0001	0.0008
Malaysia	153	10	10840	<0.0001	0.0329
Maldives	154	<10	<10	<0.0001	<0.0001
Mali	155	<10	20	<0.0001	<0.0001
Malta	156	<10	<10	<0.0001	0.0025
Marshall Islands	157	<10	<10	<0.0001	0.0001
Martinique	158	<10	<10	0.0001	0.0011
Mauritania	159	<10	20	<0.0001	<0.0001
Mauritius	160	<10	50	<0.0001	0.0225
Mayotte	161	<10	<10	<0.0001	0.0105
Mexico	162	7510	11450	0.0039	0.0059
Midway Island (U.S.)	164	<10	<10	<0.0001	<0.0001
Moldova	165	<10	<10	0.0001	0.0001
Monaco	166	<10	<10	<0.0001	0.0312
Mongolia	167	1180	1140	0.0008	0.0007
Montenegro	2647	10	140	0.0008	0.0101

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Montserrat (U.K.)	168	<10	<10	0.0012	0.0081
Morocco	169	10	470	<0.0001	0.0011
Mozambique	170	<10	240	<0.0001	0.0003
Myanmar	171	4400	15080	0.0066	0.0226
Namibia	172	<10	140	<0.0001	0.0002
Nauru	173	<10	<10	<0.0001	<0.0001
Navassa Island (U.S.)	174	<10	<10	<0.0001	0.0013
Nepal	175	4550	2930	0.0309	0.0199
Netherlands	177	<10	<10	<0.0001	<0.0001
New Caledonia (Fr.)	178	<10	50	<0.0001	0.0027
New Zealand	179	4500	1990	0.0168	0.0074
Nicaragua	180	50	470	0.0004	0.0037
Niger	181	<10	40	<0.0001	<0.0001
Nigeria	182	<10	340	<0.0001	0.0004
Niue (N.Z.)	183	N/A	N/A	N/A	N/A
Norfolk Island (Aus.)	184	<10	<10	<0.0001	0.0147
Northern Mariana Islands (U.S.)	185	<10	<10	<0.0001	0.0047
Norway	186	10	1660	<0.0001	0.0051

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Oman	187	60	420	0.0002	0.0014
Pakistan	188	750	3220	0.0009	0.0037
Palau	189	<10	30	<0.0001	0.056
Palmyra Atoll (U.S.)	190	<10	<10	<0.0001	0.0001
Panama	191	330	600	0.0044	0.008
Papua New Guinea	192	870	4850	0.0019	0.0105
Paraguay	194	<10	10	<0.0001	<0.0001
Peru	195	3020	4870	0.0023	0.0038
Philippines	196	720	23110	0.0024	0.0781
Pitcairn Islands (U.K.)	197	<10	<10	<0.0001	<0.0001
Poland	198	<10	130	<0.0001	0.0004
Portugal	199	<10	280	<0.0001	0.0032
Puerto Rico (U.S.)	200	<10	70	0.0004	0.0082
Qatar	201	<10	<10	<0.0001	<0.0001
R. B. de Venezuela	263	220	2370	0.0002	0.0026
Republic of Korea	202	<10	1670	<0.0001	0.0168
Republic of Yemen	269	<10	1080	<0.0001	0.0024
Réunion	206	<10	300	<0.0001	0.1183

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Romania	203	140	780	0.0006	0.0033
Russian Federation	204	6290	18340	0.0004	0.0011
Rwanda	205	<10	140	<0.0001	0.0053
Saba (Neth.)	75003	<10	<10	<0.0001	0.0082
Saint Barthélemy (Fr.)	75006	<10	<10	<0.0001	0.0075
Saint Helena, Ascension and Tristan da Cunha	207	<10	<10	<0.0001	0.0123
Saint Kitts and Nevis	208	<10	<10	0.0007	0.0005
Saint Lucia	209	<10	<10	0.0001	0.0016
Saint Vincent and the Grenadines	211	<10	<10	<0.0001	0.0011
Saint-Martin (Fr.)	75007	<10	<10	<0.0001	0.0125
Saint-Pierre-et-Miquelon (Fr.)	210	<10	<10	<0.0001	0.0001
Samoa	212	<10	<10	<0.0001	0.0001
San Marino	213	<10	<10	0.0004	0.0052
São Tomé and Príncipe	214	<10	<10	<0.0001	0.0007
Saudi Arabia	215	<10	330	<0.0001	0.0002
Senegal	217	<10	<10	<0.0001	<0.0001
Serbia	2648	10	240	0.0002	0.0032
Seychelles	220	<10	<10	<0.0001	0.0064

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Sierra Leone	221	<10	200	<0.0001	0.0027
Singapore	222	<10	<10	<0.0001	0.0047
Sint Eustatius (Neth.)	75004	<10	<10	<0.0001	0.0027
Sint Maarten (Neth.)	75005	<10	<10	<0.0001	0.0113
Slovak Republic	223	<10	310	<0.0001	0.0064
Slovenia	224	<10	260	0.0002	0.0126
Solomon Islands	225	460	750	0.0164	0.0265
Somalia	226	<10	230	<0.0001	0.0004
South Africa	227	<10	730	<0.0001	0.0006
South Georgia and the South Sandwich Islands (U.K.)	228	<10	10	0.0001	0.0015
South Sudan	74	<10	70	<0.0001	0.0001
Sovereign Base Areas of Akrotiri and Dhekelia (U.K.)	0	<10	<10	0.0006	0.0007
Spain	229	10	1340	<0.0001	0.0027
Sri Lanka	231	<10	40	<0.0001	0.0006
Sudan	6	<10	140	<0.0001	0.0001
Suriname	233	<10	50	<0.0001	0.0004
Svalbard and Jan Mayen (Nor.)	234	<10	10	<0.0001	0.0001

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Swaziland	235	<10	20	<0.0001	0.001
Sweden	236	<10	200	<0.0001	0.0005
Switzerland	237	<10	860	0.0001	0.0209
Syrian Arab Republic	238	30	110	0.0002	0.0006
Taiwan	147296	2930	7250	0.0811	0.2006
Tajikistan	239	5810	1410	0.0409	0.0099
Tanzania	257	10	570	<0.0001	0.0006
Thailand	240	20	3520	<0.0001	0.0068
The Bahamas	20	<10	10	<0.0001	0.0005
The Gambia	90	<10	<10	<0.0001	<0.0001
Timor-Leste	242	<10	110	0.0002	0.0076
Togo	243	<10	10	<0.0001	0.0002
Tokelau (N.Z.)	244	<10	<10	<0.0001	<0.0001
Tonga	245	<10	10	<0.0001	0.0139
Trinidad and Tobago	246	<10	20	0.0008	0.0031
Tromelin Island	247	N/A	N/A	N/A	N/A
Tunisia	248	<10	50	<0.0001	0.0003
Turkey	249	9270	4340	0.0119	0.0056

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Turkmenistan	250	140	160	0.0003	0.0003
Turks and Caicos Islands (U.K.)	251	<10	<10	<0.0001	<0.0001
Tuvalu	252	<10	<10	<0.0001	<0.0001
Uganda	253	<10	130	<0.0001	0.0005
Ukraine	254	<10	380	<0.0001	0.0006
United Arab Emirates	255	30	40	0.0005	0.0006
United Kingdom	256	<10	330	<0.0001	0.0013
United States of America	259	10710	36150	0.0011	0.0039
United States Virgin Islands (U.S.)	258	<10	<10	0.0001	0.0111
Uruguay	260	<10	10	<0.0001	0.0001
Uzbekistan	261	790	300	0.0018	0.0007
Vanuatu	262	270	430	0.022	0.0351
Vatican City	110	<10	<10	0.0001	0.0001
Vietnam	264	20	11490	0.0001	0.0351
Wake Island (U.S.)	265	<10	<10	<0.0001	<0.0001
Wallis and Futuna (Fr.)	266	<10	<10	<0.0001	0.0334
West Bank and Gaza	91	<10	30	0.0006	0.0053
Western Sahara	268	<10	<10	<0.0001	<0.0001

Country	Country code	Estimated average annual number of significant earthquake-triggered landslides	Estimated average annual number of significant rainfall-triggered landslides (1980-2018)	Estimated average annual number of significant earthquake-triggered landslides / sq. km	Estimated average annual number of significant rainfall-triggered landslides (1980-2018) / sq. km
Zambia	270	<10	40	<0.0001	<0.0001
Zimbabwe	271	<10	50	<0.0001	0.0001

Appendix C

Summary of Qualitative Landslide Hazard Classification and Landslide Seasonality

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Abyei	Very low	*Not calculated*
Afghanistan	High	Rainfall-triggered landslide hazard is highest during the months of February to May.
Albania	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of November.
Algeria	High	Rainfall-triggered landslide hazard is highest during the months of September and October.
American Samoa (U.S.)	Low	*Not calculated*
Andorra	Medium	*Not calculated*
Angola	Medium	*Not calculated*
Anguilla (U.K.)	Low	*Not calculated*
Antigua and Barbuda	Medium	*Not calculated*
Arab Republic of Egypt	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of January.
Argentina	High	Rainfall-triggered landslide hazard is highest during the months of December to March.
Armenia	High	Rainfall-triggered landslide hazard is highest during the months of May and June.
Aruba (Neth.)	Very low	*Not calculated*
Ashmore and Cartier Islands (Aus.)	NA	*Not calculated*
Australia	High	Rainfall-triggered landslide hazard is highest during the months of December to March.
Austria	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Azerbaijan	High	Rainfall-triggered landslide hazard is highest during the months of May, June and October.
Azores Islands (Por.)	High	Rainfall-triggered landslide hazard is highest during the months of December to February.
Bahrain	Very low	*Not calculated*
Baker Island (U.S.)	Very low	*Not calculated*

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Bangladesh	High	Rainfall-triggered landslide hazard is highest during the months of May to August.
Barbados	Medium	*Not calculated*
Belarus	Very low	*Not calculated*
Belgium	Low	*Not calculated*
Belize	Medium	*Not calculated*
Benin	Low	*Not calculated*
Bermuda (U.K.)	Very low	*Not calculated*
Bhutan	High	Rainfall-triggered landslide hazard is highest during the months of June to August.
Bolivia	High	Rainfall-triggered landslide hazard is highest during the months of December to March.
Bonaire (Neth.)	Very low	*Not calculated*
Bosnia and Herzegovina	High	Rainfall-triggered landslide hazard is highest during the months of September to November.
Botswana	Very low	*Not calculated*
Bouvet Island (Nor.)	Medium	*Not calculated*
Brazil	High	Rainfall-triggered landslide hazard is highest during the months of December to February.
British Indian Ocean Territory (U.K.)	Low	*Not calculated*
British Virgin Islands (U.K.)	High	Rainfall-triggered landslide hazard is highest during the months of October and November.
Brunei Darussalam	High	Rainfall-triggered landslide hazard is highest during the months of November to January.
Bulgaria	High	Rainfall-triggered landslide hazard is highest during the months of May, June, September and October.
Burkina Faso	Very low	*Not calculated*
Burundi	High	Rainfall-triggered landslide hazard is highest during the months of March and April.

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Côte d'Ivoire	Low	*Not calculated*
Cambodia	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Cameroon	High	Rainfall-triggered landslide hazard is highest during the months of August and September.
Canada	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Cape Verde	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of September.
Cayman Islands (U.K.)	Low	*Not calculated*
Central African Republic	Low	*Not calculated*
Chad	Low	*Not calculated*
Chile	High	Rainfall-triggered landslide hazard is highest during the months of May to August.
China	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Christmas Island (Aus.)	Low	*Not calculated*
Clipperton Island (Fr.)	Very low	*Not calculated*
Cocos (Keeling) Islands (Aus.)	Very low	*Not calculated*
Colombia	High	Rainfall-triggered landslide hazard is highest during the months of April, May, October and November.
Comoros	Medium	*Not calculated*
Congo	Low	*Not calculated*
Cook Islands (N.Z.)	Very low	*Not calculated*
Costa Rica	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of October.
Croatia	High	Rainfall-triggered landslide hazard is highest during the months of September to November.
Cuba	High	Rainfall-triggered landslide hazard is highest during the months of May and October.

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Curaçao (Neth.)	Very low	*Not calculated*
Cyprus	Medium	*Not calculated*
Czech Republic	High	Rainfall-triggered landslide hazard is highest during the months of May to August.
D. P. R. of Korea	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Democratic Republic of Congo	Medium	*Not calculated*
Denmark	Very low	*Not calculated*
Disputed Area[10] (Admin. 0 code 52)	Medium	*Not calculated*
Disputed Area[11] (Admin. 0 code 2)	Medium	*Not calculated*
Disputed Area[12] (Admin. 0 code 15)	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Djibouti	High	Rainfall-triggered landslide hazard is highest during the months of April, July, August and September.
Dominica	Medium	*Not calculated*
Dominican Republic	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of May.
Ecuador	High	Rainfall-triggered landslide hazard is highest during the months of January to April.
El Salvador	High	Rainfall-triggered landslide hazard is highest during the months of June to October.
Equatorial Guinea	High	Rainfall-triggered landslide hazard is highest during the months of October and November.
Eritrea	High	Rainfall-triggered landslide hazard is highest during the months of July to September.
Estonia	Very low	*Not calculated*
Ethiopia	High	Rainfall-triggered landslide hazard is highest during the months of July and August.
Falkland Islands (Malvinas)	Very low	*Not calculated*

[10] Disputed area on the border between India and China.
[11] The northern-most disputed area on the border between India and China to the east of Pakistan.
[12] The eastern-most disputed area on the border between India and China to the east of Bhutan.

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Faroe Islands (Den.)	Low	*Not calculated*
Federated States of Micronesia	Low	*Not calculated*
Fiji	High	Rainfall-triggered landslide hazard is highest during the months of January to April.
Finland	Very low	*Not calculated*
France	High	Rainfall-triggered landslide hazard is highest during the months of May, October and November.
French Guiana	Medium	*Not calculated*
French Polynesia (Fr.)	Medium	*Not calculated*
French Southern and Antarctic Lands (Fr.)	Low	*Not calculated*
FYR of Macedonia	High	Rainfall-triggered landslide hazard is highest during the months of September to November.
Gabon	Low	*Not calculated*
Georgia	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of June.
Germany	Medium	*Not calculated*
Ghana	Medium	*Not calculated*
Gibraltar (U.K.)	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of November.
Greece	High	Rainfall-triggered landslide hazard is highest during the months of November and December.
Greenland (Den.)	Very low	*Not calculated*
Grenada	Low	*Not calculated*
Guadeloupe	Low	*Not calculated*
Guam (U.S.)	Low	*Not calculated*
Guatemala	High	Rainfall-triggered landslide hazard is highest during the months of June, September and October.

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Guernsey (U.K.)	Medium	*Not calculated*
Guinea	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Guinea-Bissau	Very low	*Not calculated*
Guyana	Low	*Not calculated*
Haiti	High	*Not calculated*
Heard Island and McDonald Islands (Aus.)	High	Rainfall-triggered landslide hazard is highest during the months of March to May.
Honduras	High	Rainfall-triggered landslide hazard is highest during the months of June and October.
Hong Kong, SAR	Medium	*Not calculated*
Howland Island (U.S.)	Very low	*Not calculated*
Hungary	Low	*Not calculated*
Iceland	High	Rainfall-triggered landslide hazard is highest during the months of August to November.
India	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Indonesia	High	*Not calculated*
Iraq	High	*Not calculated*
Ireland	Medium	*Not calculated*
Islamic Republic of Iran	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of March.
Isle of Man (U.K.)	Low	*Not calculated*
Israel	Medium	*Not calculated*
Italy	High	Rainfall-triggered landslide hazard is highest during the months of September to November.
Jamaica	High	Rainfall-triggered landslide hazard is highest during the months of May, September and October.

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Japan	High	Rainfall-triggered landslide hazard is highest during the months of July to September.
Jarvis Island (U.S.)	Very low	*Not calculated*
Jersey (U.K.)	Low	*Not calculated*
Johnston Atoll (U.S.)	Very low	*Not calculated*
Jordan	High	Rainfall-triggered landslide hazard is highest during the months of December to February.
Kazakhstan	High	Rainfall-triggered landslide hazard is highest during the months of May to July.
Kenya	High	Rainfall-triggered landslide hazard is highest during the months of April, May and November.
Kingman Reef (U.S.)	NA	*Not calculated*
Kiribati	Very low	*Not calculated*
Kosovo	Medium	*Not calculated*
Kuril Islands	Low	*Not calculated*
Kuwait	Very low	*Not calculated*
Kyrgyz Republic	High	Rainfall-triggered landslide hazard is highest during the months of May to July.
Lao People's Democratic Republic	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Latvia	Very low	*Not calculated*
Lebanon	High	Rainfall-triggered landslide hazard is highest during the months of November to February.
Lesotho	Medium	*Not calculated*
Liberia	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Libya	Very low	*Not calculated*
Liechtenstein	High	Rainfall-triggered landslide hazard is highest during the months of July and August.
Lithuania	Very low	*Not calculated*

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Luxembourg	High	Rainfall-triggered landslide hazard is highest during the months of January and August.
Macau, SAR	Medium	*Not calculated*
Madagascar	High	Rainfall-triggered landslide hazard is highest during the months of December to March.
Madeira Islands (Por.)	Low	*Not calculated*
Malawi	Medium	*Not calculated*
Malaysia	High	Rainfall-triggered landslide hazard is highest during the months of November to January.
Maldives	Very low	*Not calculated*
Mali	Very low	*Not calculated*
Malta	High	Rainfall-triggered landslide hazard is highest during the months of October and November.
Marshall Islands	Very low	*Not calculated*
Martinique	Medium	*Not calculated*
Mauritania	Low	*Not calculated*
Mauritius	High	Rainfall-triggered landslide hazard is highest during the months of January to March.
Mayotte	Low	*Not calculated*
Mexico	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Midway Island (U.S.)	Very low	*Not calculated*
Moldova	Low	*Not calculated*
Monaco	High	Rainfall-triggered landslide hazard is highest during the months of October to December.
Mongolia	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Montenegro	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of November.

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Montserrat (U.K.)	Medium	*Not calculated*
Morocco	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of November.
Mozambique	Low	*Not calculated*
Myanmar	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Namibia	Low	*Not calculated*
Nauru	Very low	*Not calculated*
Navassa Island (U.S.)	Low	*Not calculated*
Nepal	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Netherlands	Very low	*Not calculated*
New Caledonia (Fr.)	Low	*Not calculated*
New Zealand	High	Rainfall-triggered landslide hazard is highest during the months of April to June.
Nicaragua	High	Rainfall-triggered landslide hazard is highest during the months of June and October.
Niger	Very low	*Not calculated*
Nigeria	Medium	*Not calculated*
Niue (N.Z.)	NA	*Not calculated*
Norfolk Island (Aus.)	Low	*Not calculated*
Northern Mariana Islands (U.S.)	Medium	*Not calculated*
Norway	High	Rainfall-triggered landslide hazard is highest during the months of September to November.
Oman	High	Rainfall-triggered landslide hazard is highest during the months of March, April and August.
Pakistan	High	Rainfall-triggered landslide hazard is highest during the months of July and August.

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Palau	Medium	*Not calculated*
Palmyra Atoll (U.S.)	Very low	*Not calculated*
Panama	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of November.
Papua New Guinea	High	Rainfall-triggered landslide hazard is highest during the months of February and March.
Paraguay	Low	*Not calculated*
Peru	High	Rainfall-triggered landslide hazard is highest during the months of December to March.
Philippines	High	*Not calculated*
Pitcairn Islands (U.K.)	Very low	*Not calculated*
Poland	High	Rainfall-triggered landslide hazard is highest during the months of May to August.
Portugal	High	Rainfall-triggered landslide hazard is predicted to be highest during the month of October.
Puerto Rico (U.S.)	High	*Not calculated*
Qatar	Very low	*Not calculated*
R. B. de Venezuela	High	Rainfall-triggered landslide hazard is highest during the months of May and June.
Réunion	Medium	*Not calculated*
Republic of Korea	High	Rainfall-triggered landslide hazard is highest during the months of July to September.
Republic of Yemen	High	Rainfall-triggered landslide hazard is highest during the months of April, July and August.
Romania	High	Rainfall-triggered landslide hazard is highest during the months of June and July.
Russian Federation	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Rwanda	High	Rainfall-triggered landslide hazard is highest during the months of March and April.
São Tomé and Príncipe	Low	*Not calculated*

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Saba (Neth.)	Medium	*Not calculated*
Saint Helena, Ascension and Tristan da Cunha (U.K.)	High	Rainfall-triggered landslide hazard is highest during the months of March to May.
Saint Kitts and Nevis	Medium	*Not calculated*
Saint Lucia	Medium	*Not calculated*
Saint Vincent and the Grenadines	Low	*Not calculated*
Saint Barthélemy (Fr.)	Medium	*Not calculated*
Saint-Martin (Fr.)	Medium	*Not calculated*
Saint-Pierre-et-Miquelon (Fr.)	Low	*Not calculated*
Samoa	Low	*Not calculated*
San Marino	Medium	*Not calculated*
Saudi Arabia	Medium	*Not calculated*
Senegal	Very low	*Not calculated*
Serbia	Medium	*Not calculated*
Seychelles	Medium	*Not calculated*
Sierra Leone	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Singapore	Low	*Not calculated*
Sint Eustatius (Neth.)	Medium	*Not calculated*
Sint Maarten (Neth.)	Medium	*Not calculated*
Slovak Republic	High	Rainfall-triggered landslide hazard is highest during the months of May to July.
Slovenia	High	Rainfall-triggered landslide hazard is highest during the months of September to November.

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Solomon Islands	High	Rainfall-triggered landslide hazard is highest during the months of January to March.
Somalia	High	Rainfall-triggered landslide hazard is highest during the months of April, May, September and October.
South Africa	Medium	*Not calculated*
South Georgia and the South Sandwich Islands (U.K.)	Low	*Not calculated*
South Sudan	Low	*Not calculated*
Sovereign Base Areas of Akrotiri and Dhekelia (U.K.)	Low	*Not calculated*
Spain	High	Rainfall-triggered landslide hazard is highest during the months of October and November.
Sri Lanka	Low	*Not calculated*
Sudan	Medium	*Not calculated*
Suriname	Low	*Not calculated*
Svalbard and Jan Mayen (Nor.)	Medium	*Not calculated*
Swaziland	High	Rainfall-triggered landslide hazard is highest during the months of November to February.
Sweden	Low	*Not calculated*
Switzerland	High	Rainfall-triggered landslide hazard is highest during the months of July and August.
Syrian Arab Republic	High	Rainfall-triggered landslide hazard is highest during the months of December to February.
Taiwan	High	Rainfall-triggered landslide hazard is highest during the months of June to September.
Tajikistan	High	Rainfall-triggered landslide hazard is highest during the months of March to June.
Tanzania	Medium	*Not calculated*
Thailand	High	Rainfall-triggered landslide hazard is highest during the months of July to September.
The Bahamas	Very low	*Not calculated*

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
The Gambia	Very low	*Not calculated*
Timor-Leste	High	Rainfall-triggered landslide hazard is highest during the months of December to March.
Togo	Medium	*Not calculated*
Tokelau (N.Z.)	Very low	*Not calculated*
Tonga	Medium	*Not calculated*
Trinidad and Tobago	High	*Not calculated*
Tromelin Island	Very low	*Not calculated*
Tunisia	Medium	*Not calculated*
Turkey	High	*Not calculated*
Turkmenistan	Low	*Not calculated*
Turks and Caicos Islands (U.K.)	Very low	*Not calculated*
Tuvalu	Very low	*Not calculated*
Uganda	High	*Not calculated*
Ukraine	High	Rainfall-triggered landslide hazard is highest during the months of June and July.
United Arab Emirates	High	Rainfall-triggered landslide hazard is highest during the months of January to April.
United Kingdom	Medium	*Not calculated*
United States of America	High	Rainfall-triggered landslide hazard is highest during the months of August to October.
United States Virgin Islands (U.S.)	High	Rainfall-triggered landslide hazard is highest during the months of September to November.
Uruguay	Very low	*Not calculated*
Uzbekistan	High	Rainfall-triggered landslide hazard is highest during the months of March and April.
Vanuatu	High	Rainfall-triggered landslide hazard is highest during the months of January to March.

Country	Admin. level 0 qualitative landslide hazard	Seasonality Text
Vatican City	Very low	*Not calculated*
Vietnam	High	Rainfall-triggered landslide hazard is highest during the months of July to September.
Wake Island (U.S.)	Very low	*Not calculated*
Wallis and Futuna (Fr.)	Medium	*Not calculated*
West Bank and Gaza	High	Rainfall-triggered landslide hazard is highest during the months of November to February.
Western Sahara	Very low	*Not calculated*
Zambia	Low	*Not calculated*
Zimbabwe	Low	*Not calculated*

Appendix D

Summary of Accompanying Data

D1 Landslide Hazard Raster Data

Raster	Description	Units
EQTL_AA.tif	Earthquake-triggered landslide hazard	Estimated average annual frequency of a significant landslide /sq. km
L_QUAL.tif	Qualitative landslide hazard	Hazard levels: 1 = Very Low 2 = Low 3 = Medium 4 = High
RFTL_AA-MEAN_1980-2018.tif	Mean Annual Rainfall-Triggered Landslide Hazard (1980-2018)	Mean annual frequency of a significant landslide / sq. km
RFTL_AA-MEDIAN_1980-2018.tif	Median Annual Rainfall-Triggered Landslide Hazard (1980-2018)	Median annual frequency of a significant landslide / sq. km

D2 Landslide Hazard Aggregated Shapefiles

Shapefile	Description	Units
ADM0_TH_Agg1.0.shp	Admin. level 0 aggregated landslide hazard levels using the 80th percentile method. Data found in the 'P80' attribute field.	Hazard levels: 1 = Very Low 2 = Low 3 = Medium 4 = High
ADM1_TH_Agg1.0.shp	Admin. level 1 aggregated landslide hazard levels using the 80th percentile method. Data found in the 'P80' attribute field.	Hazard levels: 1 = Very Low 2 = Low 3 = Medium 4 = High
ADM2_TH_Agg1.0.shp	Admin. level 2 aggregated landslide hazard levels using the 80th percentile method. Data found in the 'P80' attribute field.	Hazard levels: 1 = Very Low 2 = Low 3 = Medium 4 = High

D3 ThinkHazard! Generic Text Descriptions

Document	Description	Units
General TH!_descriptions.docx	Word document containing the four generic descriptions for very low, low, medium and high landslide hazard.	N/A

D4 Standalone Text Descriptions and Seasonality Assessment ThinkHazard Text Descriptions

Document	Description	Units
Hazard_text_and_Seasonality.csv	CSV file containing a row for each admin. 0 unit. The name, code, hazard level, hazard text and seasonality text for each admin. unit are included in separate columns.	N/A